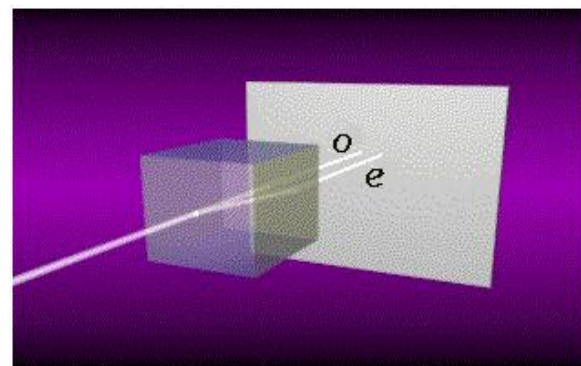


第五章 晶体光学



1. 晶体光学的电动力学基础

麦克斯韦方程组（微观）：

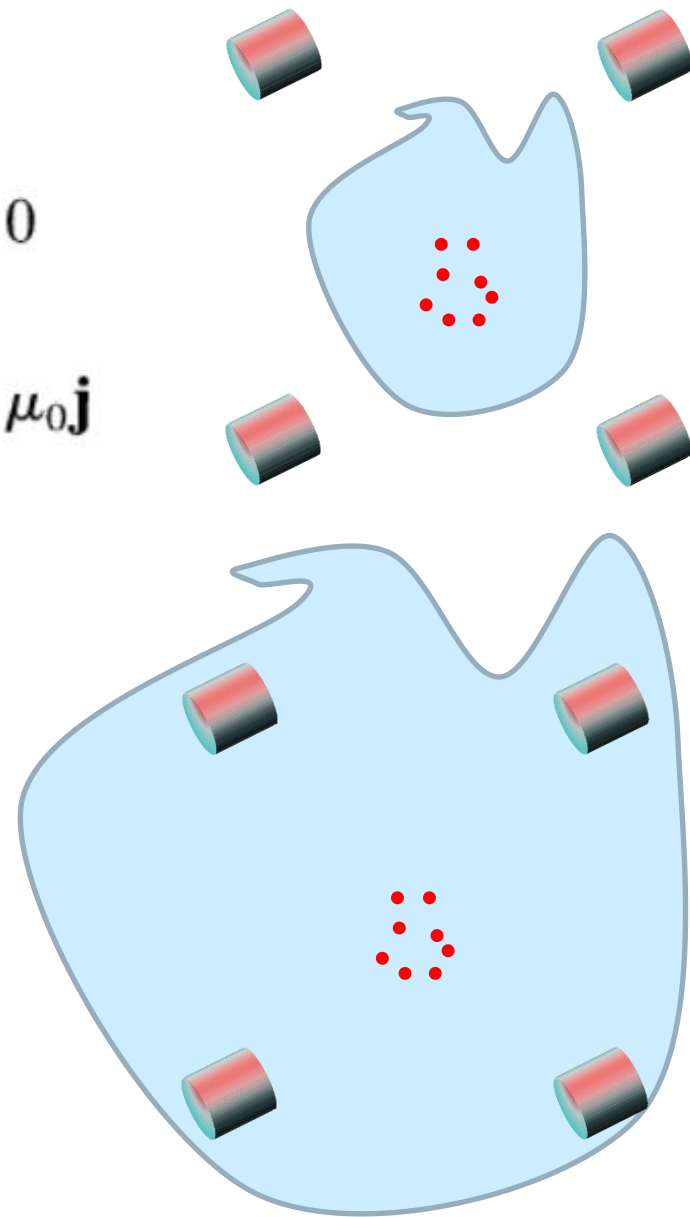
$$\nabla \cdot \mathbf{b} = 0, \quad \nabla \times \mathbf{e} + \frac{\partial \mathbf{b}}{\partial t} = 0$$

$$\nabla \cdot \mathbf{e} = \eta/\epsilon_0, \quad \nabla \times \mathbf{b} - \frac{1}{c^2} \frac{\partial \mathbf{e}}{\partial t} = \mu_0 \mathbf{j}$$

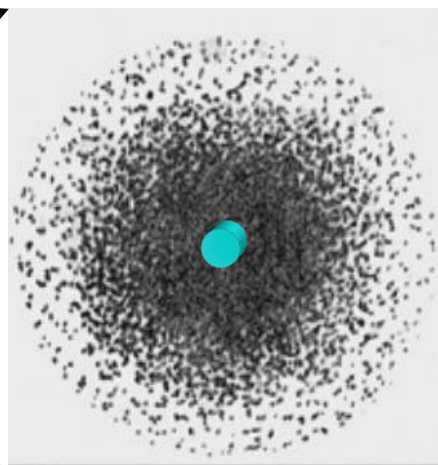
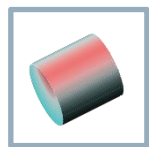
麦克斯韦方程组（宏观）：

$$\nabla \cdot \mathbf{B} = 0 \quad \nabla \times \mathbf{E} + \frac{\partial \mathbf{B}}{\partial t} = 0$$

$$\nabla \cdot \mathbf{D} = \rho \quad \nabla \times \mathbf{H} - \frac{\partial \mathbf{D}}{\partial t} = \mathbf{J}$$

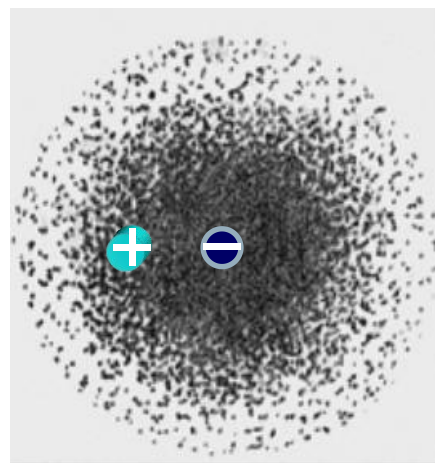
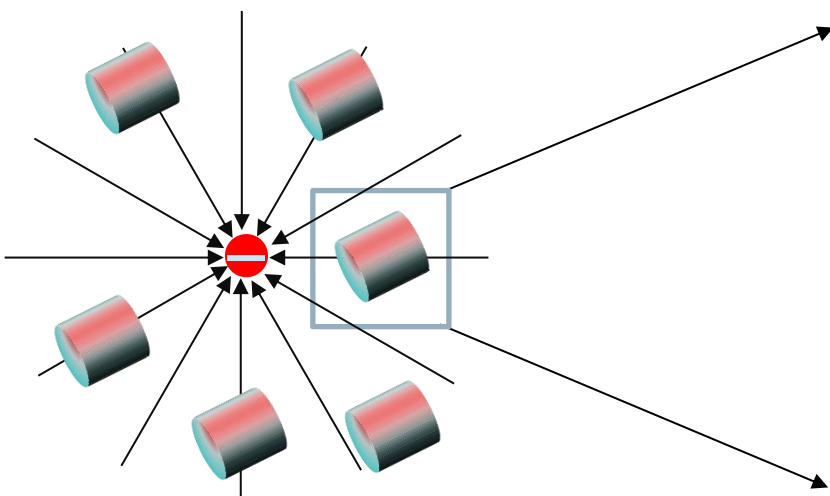


无外场下的原子



在通常状况下氢原子电子云示意图

有外场下的原子



在通常状况下氢原子电子云示意图

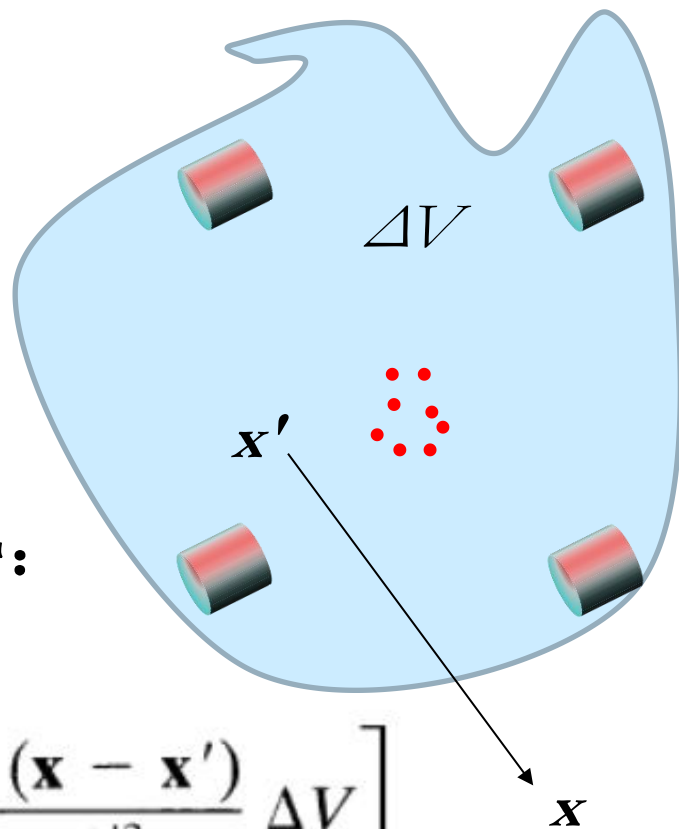
电极化

$$\Rightarrow q_i \Rightarrow P = \sum q_i$$
$$\Downarrow$$
$$P = \sum N_i q_i$$

从宏观角度，有外场情况下的电荷密度：

$$\rho(\mathbf{x}') + \sum_{i=1}^M N_i \langle e_i \rangle$$

那么， ΔV 体积内的电荷对观察点 $\mathbf{x}$ 处的电势：



$$\Delta\Phi(\mathbf{x}, \mathbf{x}') = \frac{1}{4\pi\epsilon_0} \left[\frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} \Delta V + \frac{\mathbf{P}(\mathbf{x}') \cdot (\mathbf{x} - \mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|^3} \Delta V \right]$$

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \left[\frac{\rho(\mathbf{x}')}{|\mathbf{x} - \mathbf{x}'|} + \mathbf{P}(\mathbf{x}') \cdot \nabla' \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) \right]$$

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{|\mathbf{x} - \mathbf{x}'|} [\rho(\mathbf{x}') - \nabla' \cdot \mathbf{P}(\mathbf{x}')] \text{ 有效电荷密度}$$

观察点 $\mathbf{x}$ 处的电场:

$$\mathbf{E} = -\nabla\Phi$$

$$\Phi(\mathbf{x}) = \frac{1}{4\pi\epsilon_0} \int d^3x' \frac{1}{|\mathbf{x} - \mathbf{x}'|} [\rho(\mathbf{x}') - \nabla' \cdot \mathbf{P}(\mathbf{x}')]]$$

$$\mathbf{E} = -\frac{1}{4\pi\epsilon_0} \int \nabla \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) [\rho(\mathbf{x}') - \nabla' \cdot \mathbf{P}(\mathbf{x}')] d^3x'$$

$$\nabla \cdot \mathbf{E} = -\frac{1}{4\pi\epsilon_0} \int \nabla^2 \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) [\rho(\mathbf{x}') - \nabla' \cdot \mathbf{P}(\mathbf{x}')] d^3x'$$

$$\nabla'^2 \left(\frac{1}{|\mathbf{x} - \mathbf{x}'|} \right) = -4\pi\delta(\mathbf{x} - \mathbf{x}')$$

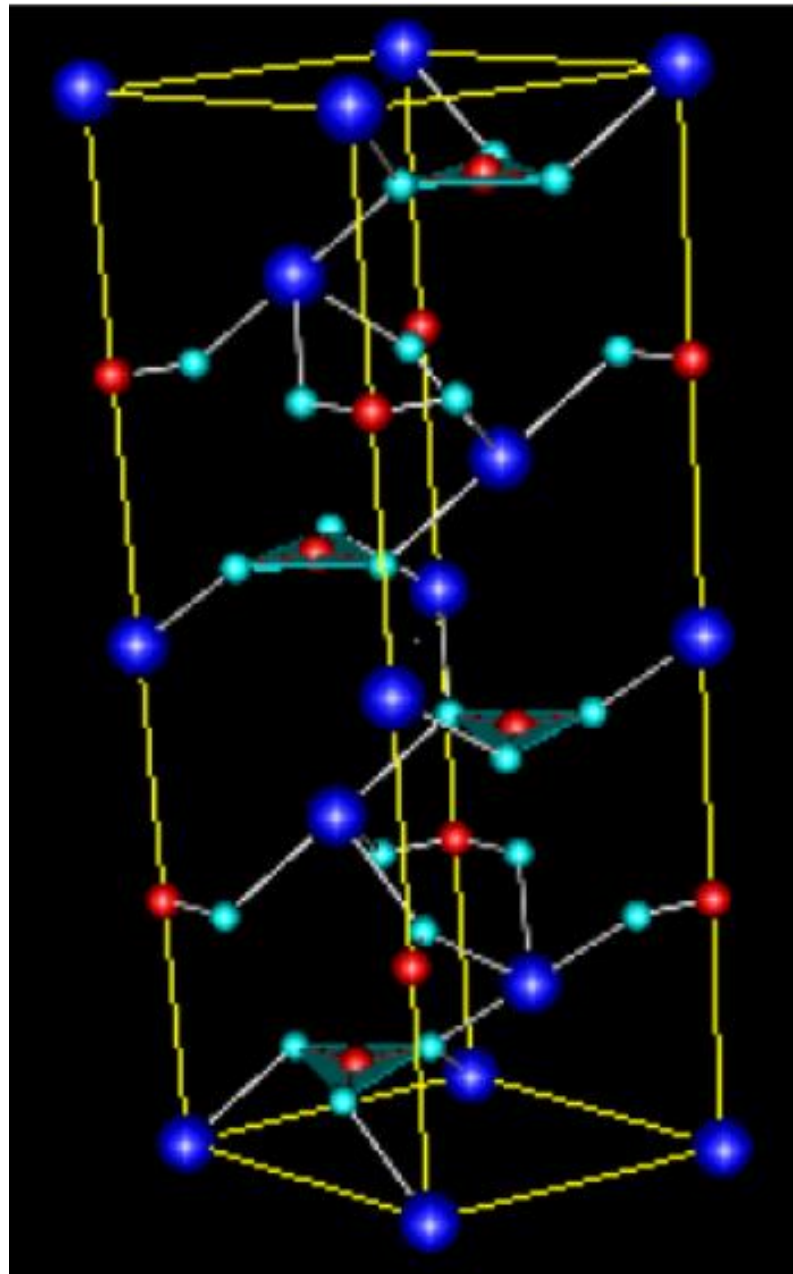
$$\nabla \cdot \mathbf{E} = \frac{\rho - \nabla \cdot \mathbf{P}}{\epsilon_0} \quad \nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho \quad \nabla \cdot \mathbf{D} = \rho$$

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P} = \epsilon_0 \mathbf{E} + \epsilon_0 \chi_e \mathbf{E} = \epsilon_0 \epsilon_r \mathbf{E} = \epsilon \mathbf{E} \quad \epsilon_r = n^2$$

$$\mathbf{D} = \vec{\epsilon} \mathbf{E}$$

$$\vec{\epsilon} = \begin{pmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{pmatrix}$$

$$\begin{pmatrix} D_1 \\ D_2 \\ D_3 \end{pmatrix} = \begin{pmatrix} \epsilon_{11} & \epsilon_{12} & \epsilon_{13} \\ \epsilon_{21} & \epsilon_{22} & \epsilon_{23} \\ \epsilon_{31} & \epsilon_{32} & \epsilon_{33} \end{pmatrix} \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix}$$



2、折射率张量、菲涅尔椭球、折射率椭球

一 各向异性介质的介电常数

物质方程: $\vec{D} = \vec{\varepsilon} \vec{E}$

其中 $\vec{\varepsilon}$ 为张量

频域

$$\vec{\varepsilon} = \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{pmatrix} \quad \vec{D} = \begin{pmatrix} D_1 \\ D_2 \\ D_3 \end{pmatrix} \quad \vec{E} = \begin{pmatrix} E_1 \\ E_2 \\ E_3 \end{pmatrix}$$

$$\begin{cases} D_1 = \varepsilon_{11}E_1 + \varepsilon_{12}E_2 + \varepsilon_{13}E_3 \\ D_2 = \varepsilon_{21}E_1 + \varepsilon_{22}E_2 + \varepsilon_{23}E_3 \\ D_3 = \varepsilon_{31}E_1 + \varepsilon_{32}E_2 + \varepsilon_{33}E_3 \end{cases} \Leftrightarrow D_i = \sum_k^3 \varepsilon_{ik} E_k$$

在各向异性介质中，能量密度的表示式仍然有效

$$\omega_e = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} \sum_{ik}^3 \varepsilon_{ik} E_k E_i$$

由麦氏方程：

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t} \quad \nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t}$$

利用公式：

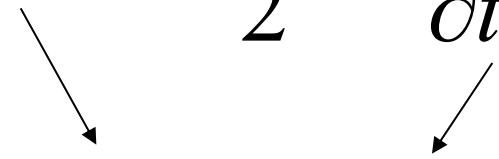
$$\nabla \cdot (\vec{E} \times \vec{H}) = -\vec{E} \cdot \nabla \times \vec{H} + \vec{H} \cdot \nabla \times \vec{E}$$

可得：

$$\nabla \cdot (\vec{E} \times \vec{H}) = -(\vec{E} \cdot \dot{\vec{D}} + \vec{H} \cdot \dot{\vec{B}})$$

代入 $D_i = \sum_k^3 \varepsilon_{ik} E_k$

$$-\nabla \cdot (\vec{E} \times \vec{H}) = \sum_{ik}^3 \varepsilon_{ik} \dot{E}_k E_i + \frac{1}{2} \frac{\partial(\mu H^2)}{\partial t}$$

$$\nabla \cdot \vec{S} = \nabla \cdot (\vec{E} \times \vec{H}) = -\left(\frac{\partial \omega_e}{\partial t} + \frac{\partial \omega_m}{\partial t}\right)$$


右边第一项：

$$\frac{\partial(\omega_e)}{\partial t} = \sum_{ik}^3 \varepsilon_{ik} \dot{E}_k E_i$$

另一方面，对电场能量密度求偏导

$$\frac{\partial \omega_e}{\partial t} = \frac{1}{2} \frac{\partial}{\partial t} \left[\sum_{ik}^3 \varepsilon_{ik} E_k E_i \right] = \frac{1}{2} \sum_{ik}^3 \varepsilon_{ik} \dot{E}_k E_i + \frac{1}{2} \sum_{ik}^3 \varepsilon_{ik} E_k \dot{E}_i$$

$$\frac{\partial(\omega_e)}{\partial t} = \sum_{ik}^3 \varepsilon_{ik} \dot{E}_k E_i$$

于是

$$\sum_{ik}^3 \varepsilon_{ik} \dot{E}_k E_i = \sum_{ik}^3 \varepsilon_{ik} E_k \dot{E}_i = \sum_{ik}^3 \varepsilon_{ki} E_i \dot{E}_k$$

$$\Rightarrow \varepsilon_{ik} = \varepsilon_{ki}$$

6个分量独立

若 ω_e 为常量:

$$\omega_e = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} \sum_{ik}^3 \varepsilon_{ik} E_k E_i$$

$$\begin{aligned} & \varepsilon_{11} E_1^2 + \varepsilon_{22} E_2^2 + \varepsilon_{33} E_3^2 + 2\varepsilon_{12} E_1 E_2 \\ & + 2\varepsilon_{13} E_1 E_3 + 2\varepsilon_{23} E_2 E_3 = \text{const} \end{aligned} \quad \varepsilon_{ik} = \varepsilon_{ki}$$

坐标变换: 新坐标沿椭球三个主轴

$$\varepsilon_1 E_1^2 + \varepsilon_2 E_2^2 + \varepsilon_3 E_3^2 = \text{const}$$

在主介电坐标系下: (坐标轴为主介电轴)

$$D_1 = \varepsilon_1 E_1 \quad D_2 = \varepsilon_2 E_2 \quad D_3 = \varepsilon_3 E_3$$

$\varepsilon_1, \varepsilon_2, \varepsilon_3$ 为主介质常数

光线椭球

$$\varepsilon_1 x^2 + \varepsilon_2 y^2 + \varepsilon_3 z^2 = 1$$

由 $\vec{D} = \vec{\varepsilon} \vec{E}$, 可得

$$\begin{cases} E_1 = \eta_{11}D_1 + \eta_{12}D_2 + \eta_{13}D_3 \\ E_2 = \eta_{21}D_1 + \eta_{22}D_2 + \eta_{23}D_3 \\ E_3 = \eta_{31}D_1 + \eta_{32}D_2 + \eta_{33}D_3 \end{cases}$$

$$\Rightarrow E_m = \sum_n^3 \eta_{mn} D_n$$

η 为 ε 的逆矩阵

$$\eta_{mn} = \frac{(\varepsilon_{ik})_{mn}}{\text{Det}(\varepsilon_{ik})}$$

代数余子式

行列式

同样可证

$$\eta_{mn} = \eta_{nm}$$

$$\omega_e = \frac{1}{2} \vec{D} \cdot \vec{E} = \frac{1}{2} \sum_{mn}^3 \eta_{mn} D_m D_n$$

取主轴坐标系可得到 D_1, D_2, D_3 为主轴的椭球球面。

$$\eta_1 D_1^2 + \eta_2 D_2^2 + \eta_3 D_3^2 = \text{const}$$

$$\left\{ \begin{array}{ll} E_1 = \eta_1 D_1 & \eta_1 = 1 / \varepsilon_1 \\ E_2 = \eta_2 D_2 & \eta_2 = 1 / \varepsilon_2 \\ E_3 = \eta_3 D_3 & \eta_3 = 1 / \varepsilon_3 \end{array} \right.$$

$$\Rightarrow \frac{D_1^2}{\varepsilon_1} + \frac{D_2^2}{\varepsilon_2} + \frac{D_3^2}{\varepsilon_3} = \textit{const}$$

$$\frac{D_1^2}{n_1^2} + \frac{D_2^2}{n_2^2} + \frac{D_3^2}{n_3^2} = \textit{const}$$

$$\frac{x^2}{n_1^2} + \frac{y^2}{n_2^2} + \frac{z^2}{n_3^2} = 1$$

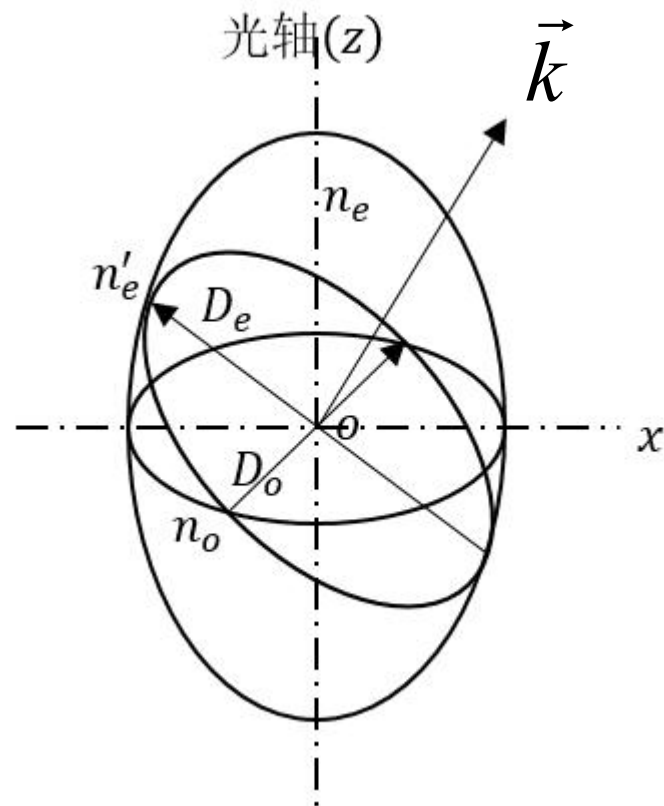
波法线椭球
折射率椭球
倒椭球

折射率的求法：

$$\frac{x^2}{n_1^2} + \frac{y^2}{n_2^2} + \frac{z^2}{n_3^2} = 1$$

$$xk_x + yk_y + zk_z = 0$$

$$r^2 = x^2 + y^2 + z^2$$



满足两个附加条件下求 r^2 的极值：
拉格朗日待定乘子法

单轴晶体 (z为光轴)

$$n_x = n_y = n_o \quad n_z = n_e$$

$$\Rightarrow \frac{x^2}{n_o^2} + \frac{y^2}{n_o^2} + \frac{z^2}{n_e^2} = 1$$

$$\left\{ \begin{array}{l} \text{正晶体: } n_e > n_o, v_e < v_o \\ \text{负晶体: } n_e < n_o, v_e > v_o \end{array} \right.$$

设 $\vec{k}$ 在 xOz 平面, 椭球—椭圆

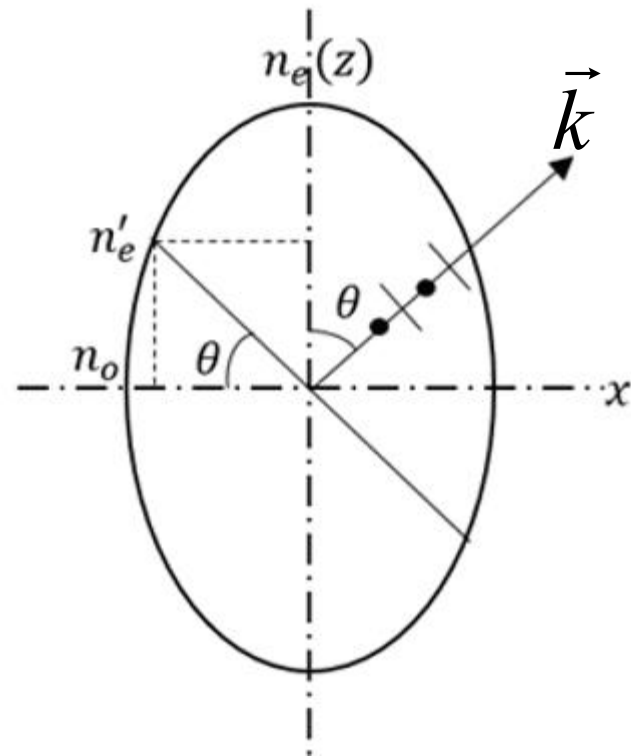
$$\frac{x^2}{n_o^2} + \frac{z^2}{n_e^2} = 1$$

A点坐标: $z = n_e' \sin \theta$

$$x = n_e' \cos \theta$$

代入椭圆方程:

$$\Rightarrow n_e' = \frac{n_o n_e}{\sqrt{n_e^2 \cos^2 \theta + n_o^2 \sin^2 \theta}}$$



θ 为 $\vec{S}$ 与光轴的夹角

单轴晶体

$$\begin{bmatrix} D_1 \\ D_2 \\ D_3 \end{bmatrix} = \begin{bmatrix} \varepsilon_o & 0 & 0 \\ 0 & \varepsilon_o & 0 \\ 0 & 0 & \varepsilon_e \end{bmatrix} \begin{bmatrix} E_1 \\ E_2 \\ E_3 \end{bmatrix}$$

1. 当光沿着z轴传播时， E_x 和 E_y 方向折射率是相同的，无双折射；当光沿x轴或y轴传播时，有双折射
2. 双折射的本质：矢量光束在经过晶体时，不同方向上的电场分量，具有不同的折射率（影响光的位相）。需要两个要素：分子的极化（即电偶极子的激发）和分子的规则排列。
3. 广义双折射：矢量光束在经过物质或材料时，不同方向上的电场分量，具有不同的光学响应（光的位相、振幅、偏振）。

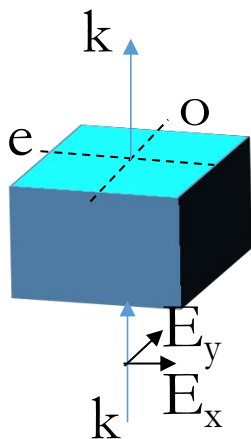
如何激发电偶极子？ 如何排列电偶极子？

超表面器件的双折射效应

$$E_{trans} = E_x e^{ikh n_e} \mathbf{e}_x + E_y e^{ikh n_o} \mathbf{e}_y$$

$$= E_x \mathbf{e}_x + E_y e^{ikh(n_o - n_e)} \mathbf{e}_y$$

传统晶体



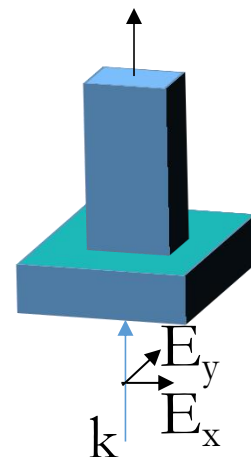
$$E_{in} = E_x \mathbf{e}_x + E_y \mathbf{e}_y$$

$$E_{trans} = E_x t_x \mathbf{e}_x + E_y t_y \mathbf{e}_y$$

$$= E_x A_x e^{i\varphi_x} \mathbf{e}_x + E_y A_y e^{i\varphi_y} \mathbf{e}_y$$

$$= E_x \mathbf{e}_x + E_y \frac{A_y}{A_x} e^{i(\varphi_y - \varphi_x)} \mathbf{e}_y$$

超表面器件



$$E_{in} = E_x \mathbf{e}_x + E_y \mathbf{e}_y$$

3、偏振态的矩阵表示

- 每一种偏振态的电矢量都可用两个正交分量表示

$$\begin{cases} E_x = A_x \cos(kz - \omega t + \varphi_{0x}) \\ E_y = A_y \cos(kz - \omega t + \varphi_{0y}) \end{cases}$$

- 也可以用复振幅表示为

$$\begin{cases} \tilde{E}_x = A_x e^{i(kz - \omega t + \varphi_{0x})} = A_x e^{i(kz + \varphi_{0x})} e^{-i\omega t} = A_x e^{i\varphi_x(z)} e^{-i\omega t} \\ \tilde{E}_y = A_y e^{i(kz - \omega t + \varphi_{0y})} = A_y e^{i(kz + \varphi_{0y})} e^{-i\omega t} = A_y e^{i\varphi_y(z)} e^{-i\omega t} \end{cases} \Rightarrow \begin{cases} \tilde{E}_x = A_x e^{i\varphi_x(z)} \\ \tilde{E}_y = A_y e^{i\varphi_y(z)} \end{cases}$$

- 用 2×1 矩阵表示为 $E = \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} = \begin{bmatrix} A_x e^{i\varphi_x(z)} \\ A_y e^{i\varphi_y(z)} \end{bmatrix}$ 琼斯矢量

琼斯矢量的矩阵表示

■ 偏振光的琼斯矢量
$$\mathbf{E} = \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} = \begin{bmatrix} A_x e^{i\varphi_x(z)} \\ A_y e^{i\varphi_y(z)} \end{bmatrix}$$

• 偏振光的强度
$$I = |\tilde{E}_x|^2 + |\tilde{E}_y|^2 = A_x^2 + A_y^2$$

• 归一化的琼斯矢量
$$\mathbf{E} = \frac{1}{\sqrt{A_x^2 + A_y^2}} \begin{bmatrix} A_x e^{i\varphi_x(z)} \\ A_y e^{i\varphi_y(z)} \end{bmatrix}$$

• 偏振态取决于两分量的相位差和振幅之比

$$\mathbf{E} = A_x e^{i\varphi_x} \begin{bmatrix} 1 \\ (A_y / A_x) e^{i(\varphi_y - \varphi_x)} \end{bmatrix} = A_x e^{i\varphi_x} \begin{bmatrix} 1 \\ \gamma e^{i\varphi} \end{bmatrix} \quad \gamma = \frac{A_y}{A_x}$$

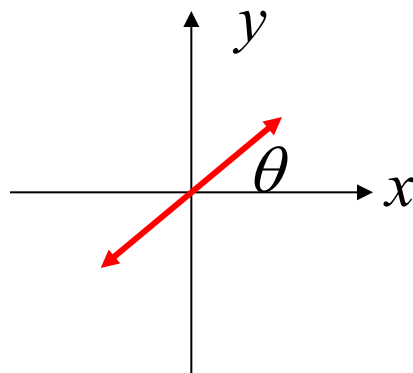
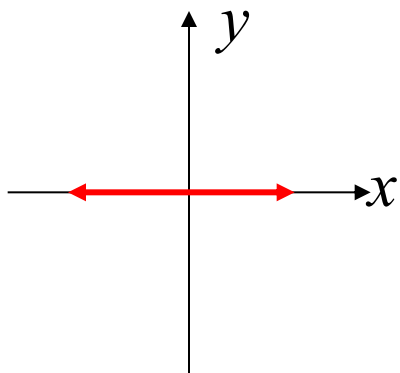
平面偏振光的琼斯矢量

- 电矢量沿x轴的线偏光的归一化琼斯矢量

$$\mathbf{E} = \frac{1}{A_x} \begin{bmatrix} A_x e^{i\varphi_x} \\ 0 \end{bmatrix} = e^{i\varphi_x} \begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}$$

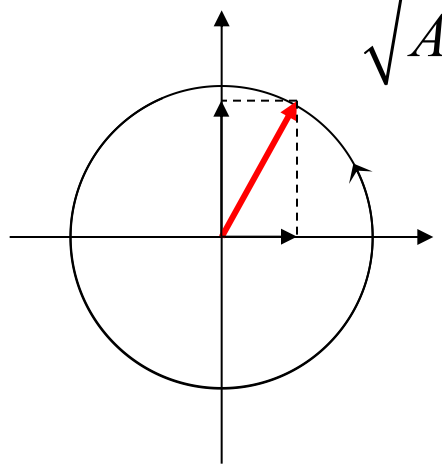
- 光矢量与x轴夹角为 θ 的线偏光的琼斯矢量

$$\mathbf{E} = \frac{1}{\sqrt{A_x^2 + A_y^2}} \begin{bmatrix} A_x e^{i\varphi_x(z)} \\ A_y e^{i\varphi_y(z)} \end{bmatrix} = \frac{e^{i\varphi_x}}{A \sqrt{\cos^2 \theta + \sin^2 \theta}} \begin{bmatrix} A \cos \theta \\ A \sin \theta \end{bmatrix} = e^{i\varphi_x} \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix} \quad \mathbf{E} = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$$

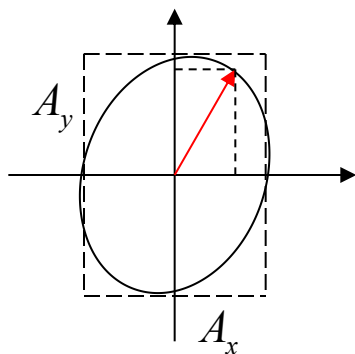


圆偏振光的琼斯矢量

- 在复振幅表达式中，相位值大表示滞后


$$\mathbf{E} = \frac{1}{\sqrt{A_x^2 + A_y^2}} \begin{bmatrix} A_x e^{i\varphi_x} \\ A_y e^{i\varphi_y} \end{bmatrix} = \frac{1}{\sqrt{2}A} \begin{bmatrix} A e^{i\varphi_x} \\ A e^{i\varphi_x \pm \frac{\pi}{2}} \end{bmatrix} = \frac{e^{i\varphi_x}}{\sqrt{2}} \begin{bmatrix} 1 \\ e^{\pm i\frac{\pi}{2}} \end{bmatrix}$$
$$\mathbf{E} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ \pm i \end{bmatrix} \quad +i \text{左旋}, -i \text{右旋}$$

椭圆偏振光的琼斯矢量



$$\mathbf{E} = \frac{1}{\sqrt{A_x^2 + A_y^2}} \begin{bmatrix} A_x e^{i\varphi_x} \\ A_y e^{i\varphi_y} \end{bmatrix} \quad \gamma = \frac{A_y}{A_x}$$

$$\mathbf{E} = \frac{1}{\sqrt{1 + \gamma^2}} \begin{bmatrix} 1 \\ \gamma e^{i\varphi} \end{bmatrix}$$

- 两分量的相位差 φ 处于I、II象限时，为左旋；处于III、VI象限时，为右旋。
- 当 $\varphi = \pm \pi/2$ ，且 $\gamma \neq 1$ 时，为正椭圆，其长短轴与坐标轴平行。

正交偏振

■ 两列光波的琼斯矢量

$$\mathbf{E}_1 = \begin{bmatrix} \tilde{E}_{1x} \\ \tilde{E}_{1y} \end{bmatrix} \quad \mathbf{E}_2 = \begin{bmatrix} \tilde{E}_{2x} \\ \tilde{E}_{2y} \end{bmatrix}$$

- 如果 $\mathbf{E}_1 \cdot \mathbf{E}_2^* = \tilde{E}_{1x}\tilde{E}_{2x}^* + \tilde{E}_{1y}\tilde{E}_{2y}^* = 0$
- 则这两列光波是正交偏振的

正交偏振的物理意义

■ 两列平面偏振光 $\mathbf{E}_1 = \begin{bmatrix} \cos \theta_1 \\ \sin \theta_1 \end{bmatrix}$ $\mathbf{E}_2 = \begin{bmatrix} \cos \theta_2 \\ \sin \theta_2 \end{bmatrix}$

$$\mathbf{E}_1 \cdot \mathbf{E}_2^* = \begin{bmatrix} \cos \theta_2 & \sin \theta_2 \end{bmatrix} \begin{bmatrix} \cos \theta_1 \\ \sin \theta_1 \end{bmatrix} = \cos \theta_1 \cos \theta_2 + \sin \theta_1 \sin \theta_2$$

$$= \cos(\theta_1 - \theta_2) = 0$$

$$\theta_2 = \theta_1 + \frac{\pi}{2}$$

• 两列正交的平面偏振光 $\mathbf{E}_1 = \begin{bmatrix} \cos \theta \\ \sin \theta \end{bmatrix}$ $\mathbf{E}_2 = \begin{bmatrix} -\sin \theta \\ \cos \theta \end{bmatrix}$

正交的圆偏振光

- 左旋圆偏振光 $\mathbf{E}_L = \begin{bmatrix} 1 \\ i \end{bmatrix}$

- 右旋圆偏振光 $\mathbf{E}_R = \begin{bmatrix} 1 \\ -i \end{bmatrix}$

$$\mathbf{E}_L \cdot \mathbf{E}_R^* = \begin{bmatrix} 1 & i \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = 1 - 1 = 0$$

- 是相互正交的

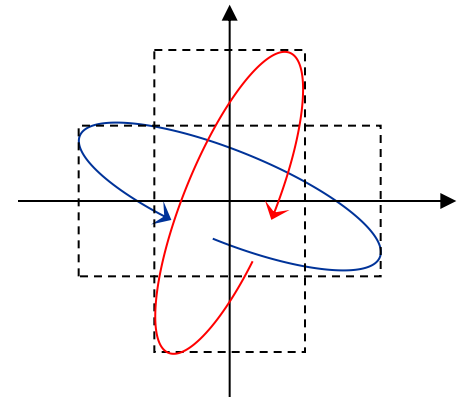
正交的椭圆偏振光

■ 两列椭圆偏振光 $\mathbf{E}_1 = \frac{1}{\sqrt{1+\gamma_1^2}} \begin{bmatrix} 1 \\ \gamma_1 e^{i\varphi_1} \end{bmatrix}$ $\mathbf{E}_2 = \frac{1}{\sqrt{1+\gamma_2^2}} \begin{bmatrix} 1 \\ \gamma_2 e^{i\varphi_2} \end{bmatrix}$

$$\mathbf{E}_1 \cdot \mathbf{E}_2^* = \tilde{E}_{1x} \tilde{E}_{2x}^* + \tilde{E}_{1y} \tilde{E}_{2y}^* = 1 + \gamma_1 \gamma_2 e^{i\varphi_1} e^{-i\varphi_2} = 0 \quad \gamma_1 \gamma_2 e^{i(\varphi_1 - \varphi_2)} = -1$$

$$\begin{cases} \gamma_2 \gamma_1 = 1 \\ \varphi_1 - \varphi_2 = \pm \pi \end{cases} \quad \gamma_1 = \frac{A_{1y}}{A_{1x}} \quad \gamma_2 = \frac{A_{2y}}{A_{2x}} \quad \gamma_1 \gamma_2 = \frac{A_{1y}}{A_{1x}} \frac{A_{2y}}{A_{2x}}$$

- 旋向相反，长短轴成比例



偏振态的正交分解

- 任意的偏振光 $\mathbf{E} = \begin{bmatrix} A \\ B \end{bmatrix}$
- 分解为正交的平面偏振光 $\begin{bmatrix} A \\ B \end{bmatrix} = A \begin{bmatrix} 1 \\ 0 \end{bmatrix} + B \begin{bmatrix} 0 \\ 1 \end{bmatrix}$
- 分解为正交的圆偏振光 $\begin{bmatrix} A \\ B \end{bmatrix} = \frac{A+iB}{2} \begin{bmatrix} 1 \\ -i \end{bmatrix} + \frac{A-iB}{2} \begin{bmatrix} 1 \\ i \end{bmatrix}$
- 分解为正交的椭圆偏振光？

琼斯矩阵

- 经过偏振元件后，光的偏振态改变

$$\mathbf{E} = \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} \quad \mathbf{E}' = \begin{bmatrix} \tilde{E}'_x \\ \tilde{E}'_y \end{bmatrix} \quad \mathbf{E}' = \mathbf{P}\mathbf{E} \quad \text{琼斯矩阵}$$

- 偏振元件应该是一个 2×2 矩阵 $\mathbf{P} = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix}$

$$\mathbf{E}' = \begin{bmatrix} \tilde{E}'_x \\ \tilde{E}'_y \end{bmatrix} = \mathbf{P}\mathbf{E} = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} = \begin{bmatrix} p_{11}\tilde{E}_x + p_{12}\tilde{E}_y \\ p_{21}\tilde{E}_x + p_{22}\tilde{E}_y \end{bmatrix} \quad \begin{cases} \tilde{E}'_x = p_{11}\tilde{E}_x + p_{12}\tilde{E}_y \\ \tilde{E}'_y = p_{21}\tilde{E}_x + p_{22}\tilde{E}_y \end{cases}$$

- 依次经过多个偏振元件 $\mathbf{E}' = \mathbf{P}_n \cdots \mathbf{P}_2 \mathbf{P}_1 \mathbf{E}$

典型偏振元件的琼斯矩阵

■ $\frac{1}{2}$ 波片，快轴沿x方向

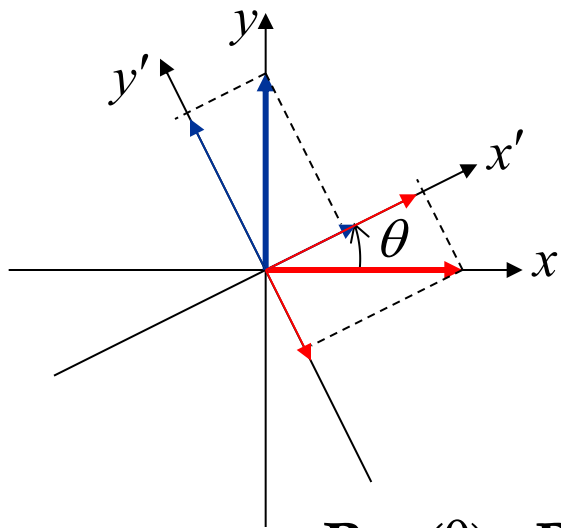
$$\mathbf{E} = \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} \quad \mathbf{E}' = \begin{bmatrix} e^{i\varphi} \tilde{E}_x \\ e^{i\varphi} e^{i\pi} \tilde{E}_y \end{bmatrix} = e^{i\varphi} \begin{bmatrix} \tilde{E}_x \\ e^{i\pi} \tilde{E}_y \end{bmatrix} \quad \mathbf{E}' = \begin{bmatrix} \tilde{E}_x \\ e^{i\pi} \tilde{E}_y \end{bmatrix}$$

$$\begin{bmatrix} \tilde{E}_x \\ e^{i\pi} \tilde{E}_y \end{bmatrix} = \begin{bmatrix} p_{11} & p_{12} \\ p_{21} & p_{22} \end{bmatrix} \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} = \begin{bmatrix} p_{11}\tilde{E}_x + p_{12}\tilde{E}_y \\ p_{21}\tilde{E}_x + p_{22}\tilde{E}_y \end{bmatrix} \quad \mathbf{P}_{+\lambda/2} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

• $\frac{1}{2}$ 波片，快轴沿y方向

$$\mathbf{E}' = \begin{bmatrix} e^{i\pi} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} \quad \mathbf{P}_{-\lambda/2} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$$

- $\frac{1}{2}$ 波片，快轴与 x 方向夹角 θ



$$\begin{cases} x' = x \cos \theta + y \sin \theta \\ y' = -x \sin \theta + y \cos \theta \end{cases}$$

旋转操作的Jones矩阵:

$$\mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

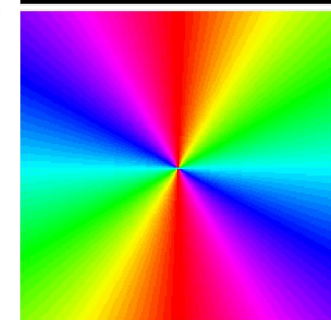
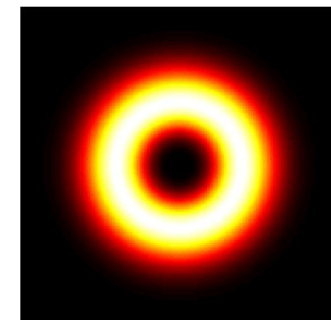
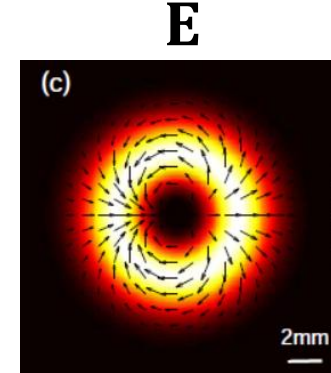
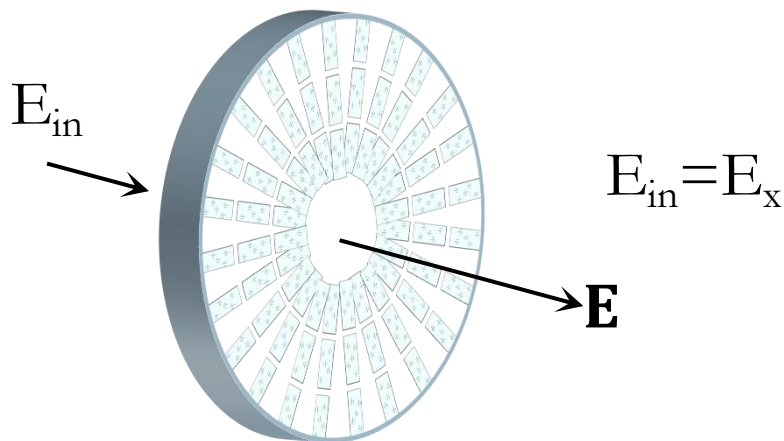
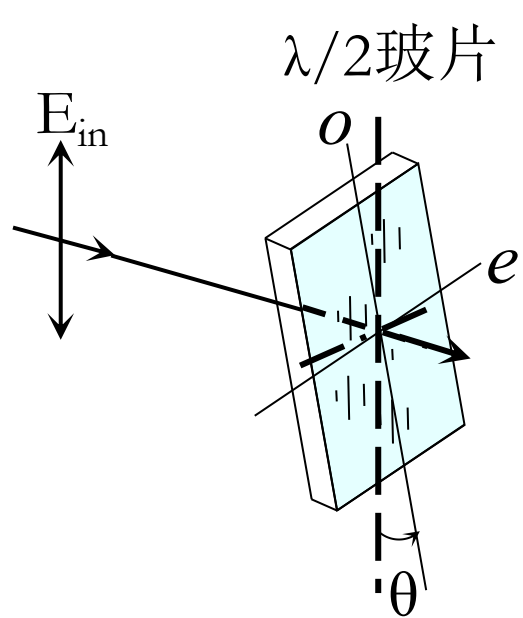
$$\mathbf{P}_{+\lambda/2} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\mathbf{R}(-\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\mathbf{P}_{\lambda/2}(\theta) = \mathbf{R}(-\theta) \mathbf{P}_{\lambda/2}(0) \mathbf{R}(\theta)$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix}$$

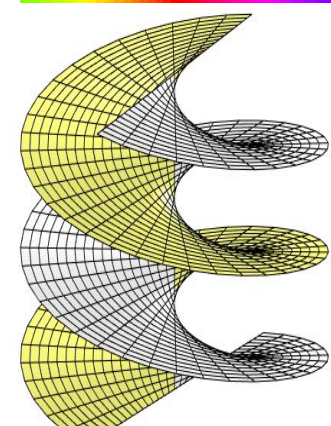
$$= \begin{bmatrix} \cos^2 \theta - \sin^2 \theta & 2 \sin \theta \cos \theta \\ 2 \cos \theta \sin \theta & \sin^2 \theta - \cos^2 \theta \end{bmatrix} = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$



$$\mathbf{E} = \mathbf{P}_{\frac{\lambda}{2}}(\theta) \mathbf{E}_x = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} \cos 2\theta \\ \sin 2\theta \end{bmatrix} \quad E_{in} = E_{cp}$$

$$\mathbf{E} = \mathbf{P}_{\frac{\lambda}{2}}(\theta) \mathbf{E}_y = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} \sin 2\theta \\ -\cos 2\theta \end{bmatrix}$$

$$\begin{aligned} \mathbf{E} = \mathbf{P}_{\frac{\lambda}{2}}(\theta) \mathbf{E}_{cp} &= \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix} \begin{bmatrix} 1 \\ \pm i \end{bmatrix} = \begin{bmatrix} \cos 2\theta \pm i \sin 2\theta \\ \sin 2\theta \mp i \cos 2\theta \end{bmatrix} \\ &= e^{\pm 2i\theta} \begin{bmatrix} 1 \\ \mp i \end{bmatrix} \end{aligned}$$



- $\frac{1}{4}$ 波片，快轴沿 x 方向

$$\mathbf{E} = \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} \quad \mathbf{E}' = \begin{bmatrix} \tilde{E}_x \\ e^{i\pi/2} \tilde{E}_y \end{bmatrix} \quad \mathbf{P}_{+\lambda/4} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

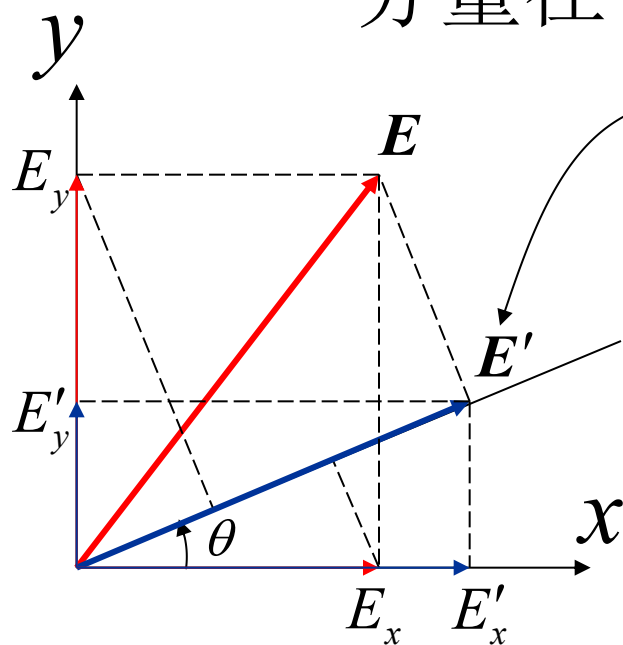
- $\frac{1}{4}$ 波片，快轴与 x 方向夹角 θ

$$\begin{aligned} \mathbf{P}_{\lambda/4}(\theta) &= \mathbf{R}(-\theta) \mathbf{P}_{\lambda/4}(0) \mathbf{R}(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + i \sin^2 \theta & \sin \theta \cos \theta - i \sin \theta \cos \theta \\ \sin \theta \cos \theta - i \sin \theta \cos \theta & \sin^2 \theta + i \cos^2 \theta \end{bmatrix} \end{aligned}$$

线起偏器

■ 透振方向与x轴夹角 θ

出射光矢量的大小等于入射光两正交分量在偏振片透振方向上的投影之和



$$E' = E_x \cos \theta + E_y \sin \theta$$

$$\begin{cases} E'_x = (E_x \cos \theta + E_y \sin \theta) \cos \theta \\ E'_y = (E_x \cos \theta + E_y \sin \theta) \sin \theta \end{cases}$$

$$\begin{cases} E'_x = E_x \cos^2 \theta + E_y \sin \theta \cos \theta \\ E'_y = E_x \sin \theta \cos \theta + E_y \sin^2 \theta \end{cases}$$

线起偏器的琼斯矩阵

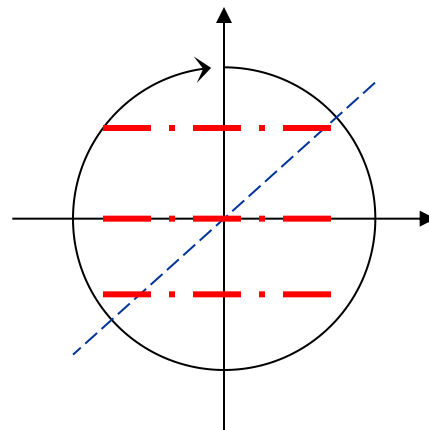
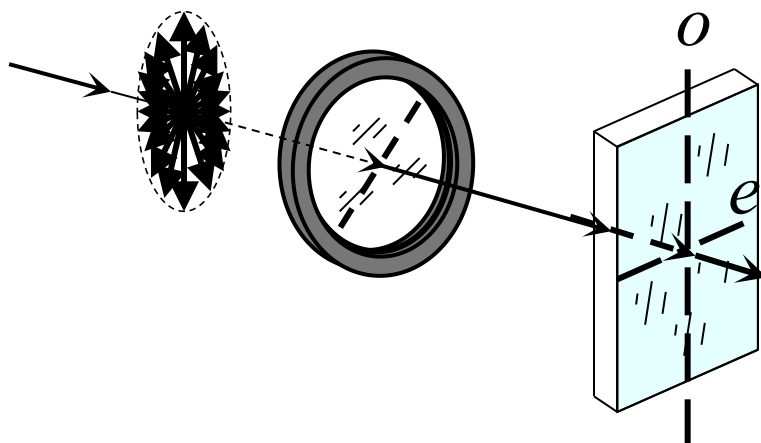
$$\begin{cases} \tilde{E}'_x = \tilde{E}_x \cos^2 \theta + \tilde{E}_y \sin \theta \cos \theta \\ \tilde{E}'_y = \tilde{E}_x \sin \theta \cos \theta + \tilde{E}_y \sin^2 \theta \end{cases}$$

$$\begin{bmatrix} \tilde{E}'_x \\ \tilde{E}'_y \end{bmatrix} = \begin{bmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{bmatrix} \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix}$$

$$\mathbf{P} = \begin{bmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{bmatrix}$$

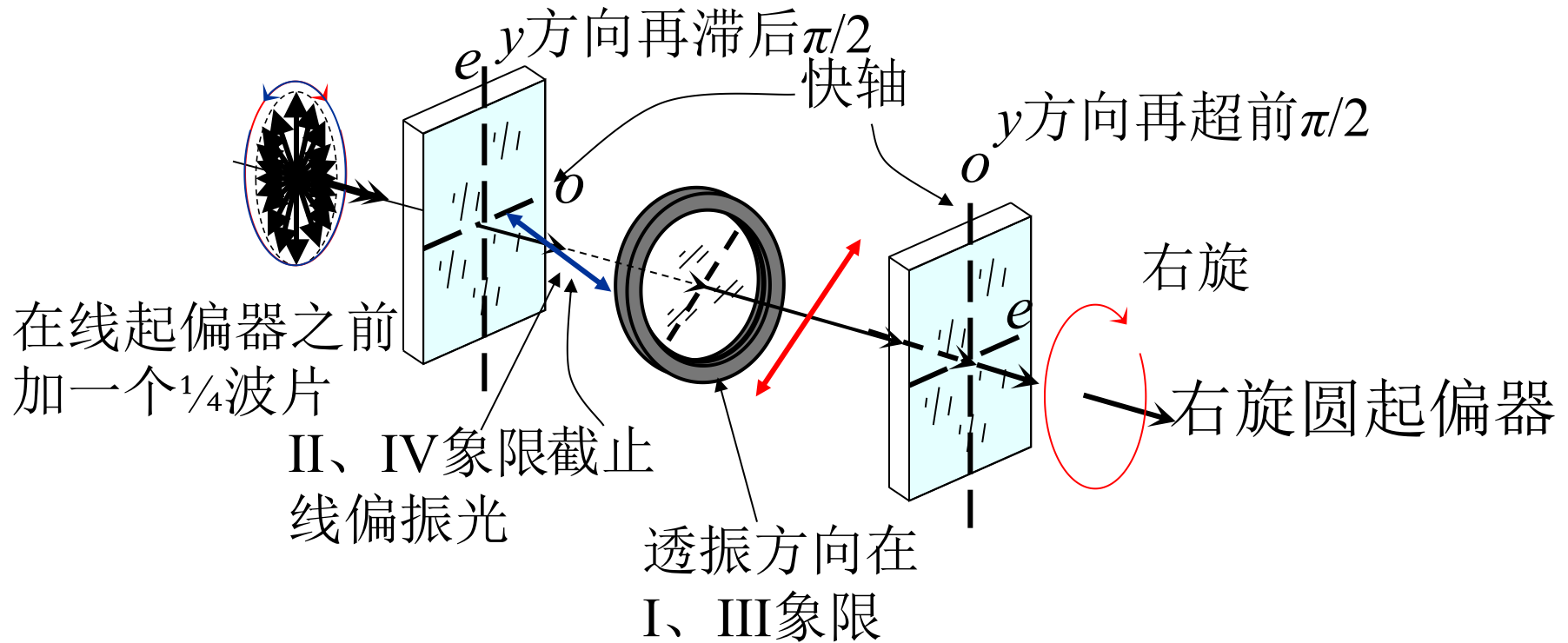
圆起偏器

- 光依次通过一个偏振片和一个 $\frac{1}{4}$ 波片，可以变为圆偏振光，圆起偏器。
- 偏振片的透振方向与波片的光轴成 $\pi/4$ 角



单向圆起偏器

- 左旋光和右旋光是正交的
- 所以圆起偏器要能阻挡反向的圆偏振光



圆起偏器的琼斯矩阵

- 两个正交 $1/4$ 波片之间插入一个偏振片

$$\mathbf{P}_{-\lambda/4} = \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} \quad \mathbf{P} = \begin{bmatrix} \cos^2 \frac{\pi}{4} & \sin \frac{\pi}{4} \cos \frac{\pi}{4} \\ \sin \frac{\pi}{4} \cos \frac{\pi}{4} & \sin^2 \frac{\pi}{4} \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \quad \mathbf{P}_{\lambda/4} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$$\text{琼斯矩阵} \quad \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \frac{1}{2} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -i \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \begin{bmatrix} \tilde{E}_x \\ \tilde{E}_y \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \tilde{E}_x + i\tilde{E}_y \\ -i\tilde{E}_x + \tilde{E}_y \end{bmatrix} = \frac{1}{2} \begin{bmatrix} \tilde{E}_x + i\tilde{E}_y \\ -i(\tilde{E}_x + i\tilde{E}_y) \end{bmatrix} = \frac{\tilde{E}_x + i\tilde{E}_y}{2} \begin{bmatrix} 1 \\ e^{-i\pi/2} \end{bmatrix}$$

$$\frac{1}{2} \begin{bmatrix} 1 & i \\ -i & 1 \end{bmatrix} \begin{bmatrix} 1 \\ i \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1-1 \\ -i+i \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Jones矢量:

x偏振光 $\begin{bmatrix} 1 \\ 0 \end{bmatrix}$ y偏振光 $\begin{bmatrix} 0 \\ 1 \end{bmatrix}$ 圆偏振光 $\begin{bmatrix} 1 \\ \pm i \end{bmatrix}$ 柱矢量偏振光 $\begin{bmatrix} \cos(n\theta) \\ \sin(n\theta) \end{bmatrix}$

Jones矩阵:

$$\mathbf{P}_{+\lambda/2} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \qquad \mathbf{P}_{+\lambda/4} = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$$

$$\mathbf{P}_{\lambda/2}(\theta) = \begin{bmatrix} \cos 2\theta & \sin 2\theta \\ \sin 2\theta & -\cos 2\theta \end{bmatrix}$$

$$\mathbf{P}_{\lambda/4}(\theta) = \begin{bmatrix} \cos^2 \theta + i \sin^2 \theta & \sin \theta \cos \theta - i \sin \theta \cos \theta \\ \sin \theta \cos \theta - i \sin \theta \cos \theta & \sin^2 \theta + i \cos^2 \theta \end{bmatrix}$$

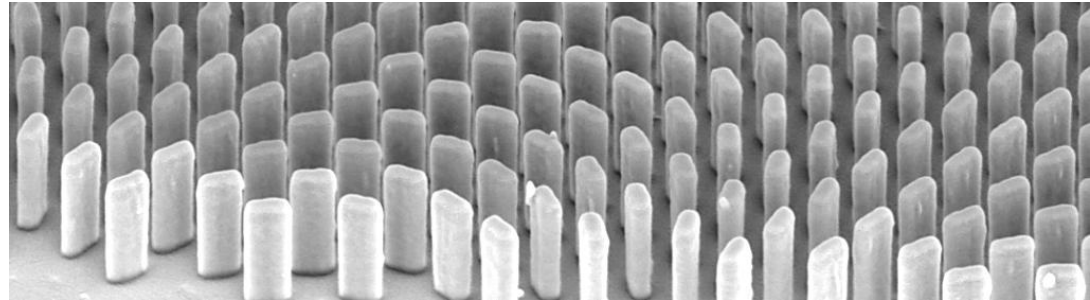
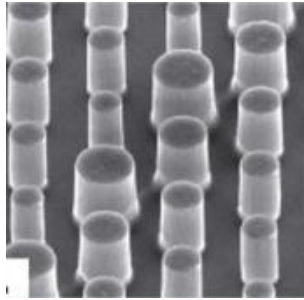
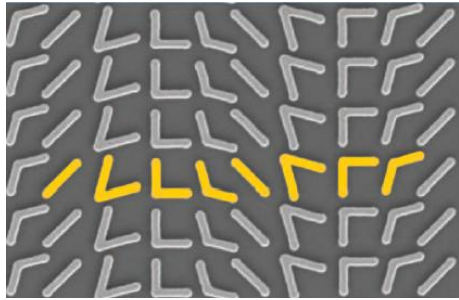
$$\mathbf{P} = \begin{bmatrix} \cos^2 \theta & \sin \theta \cos \theta \\ \sin \theta \cos \theta & \sin^2 \theta \end{bmatrix} \quad \text{线偏振片}$$

§ 4、微纳光学器件中的双折射现象

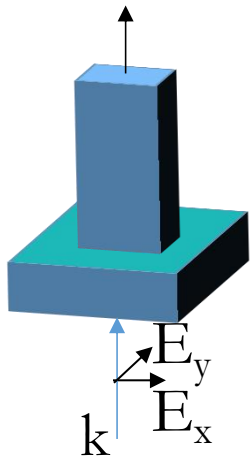
——超表面偏振光学

1. 超表面简介
2. 超表面工作原理
3. 效率问题
4. 应用（透镜，光栅，全息）
5. 计算模拟（时域有限差分法）

4.1 超表面介绍



超表面器件



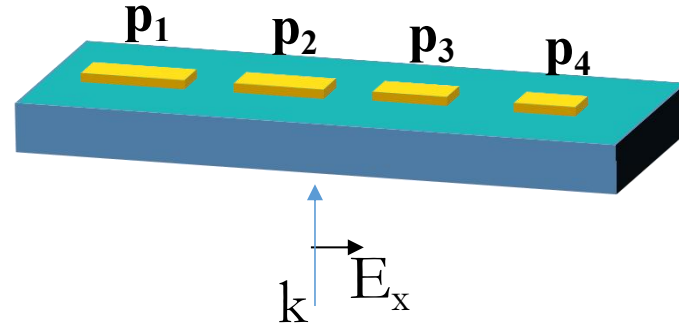
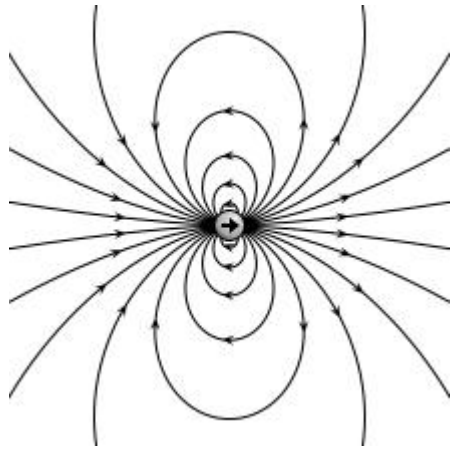
$$E_x \mathbf{e}_x + E_y \frac{A_y}{A_x} e^{i(\varphi_y - \varphi_x)} \mathbf{e}_y$$

特点：

1. 高度：亚波长、平板性
2. 像素（周期）：亚波长
3. 结构：空间上和结构上均变化
(形状、大小、旋转)
4. 材料：金属、透明电介质
5. 功能：**位相**、振幅、偏振、波长调控

4.2 超表面原理

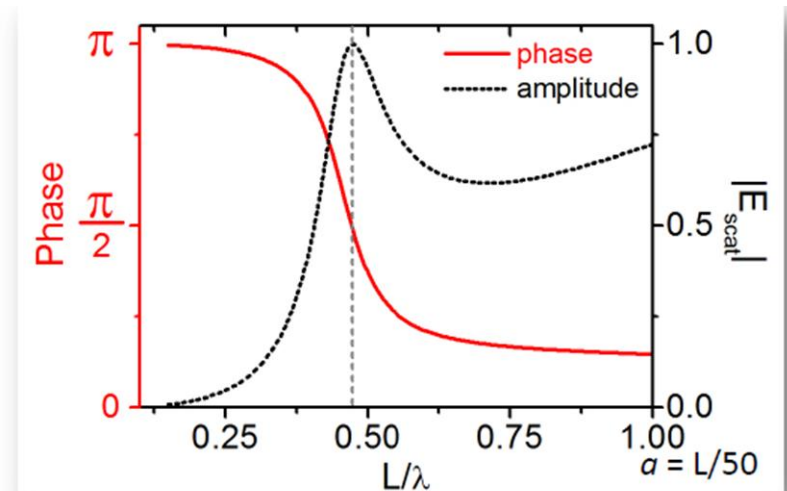
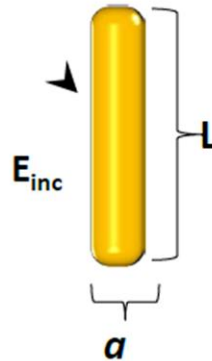
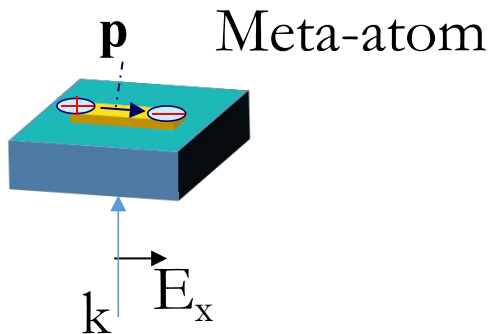
1. 电偶极子震荡



$$\mathbf{E}_{scat}(\mathbf{x}) = \sum_{i=1}^N \frac{3\mathbf{n}(\mathbf{p}_i \cdot \mathbf{n}) - \mathbf{p}_i}{4\pi\epsilon_0|\mathbf{x} - \mathbf{x}_i|^3}$$

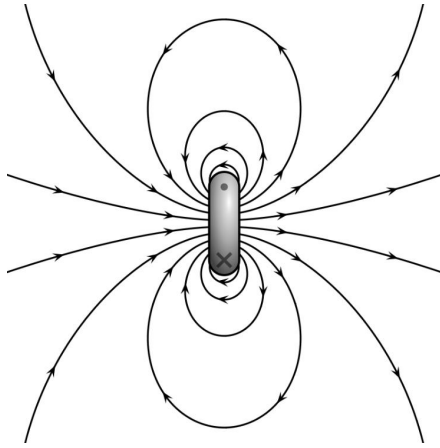
电偶极子辐射的电场:

$$\mathbf{E}(\mathbf{x}) = \frac{3\mathbf{n}(\mathbf{p} \cdot \mathbf{n}) - \mathbf{p}}{4\pi\epsilon_0|\mathbf{x} - \mathbf{x}_0|^3}$$

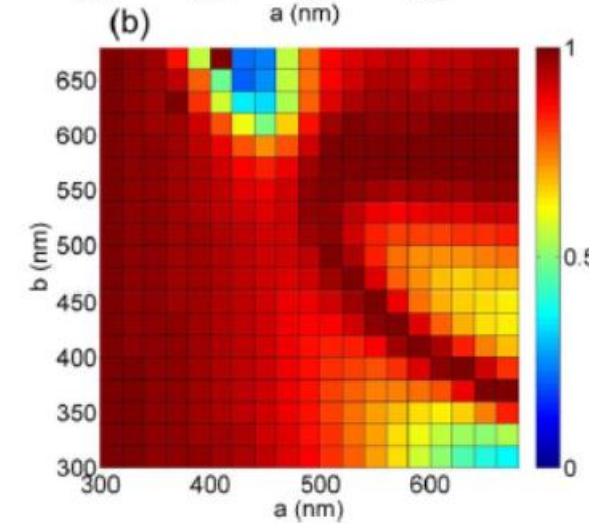
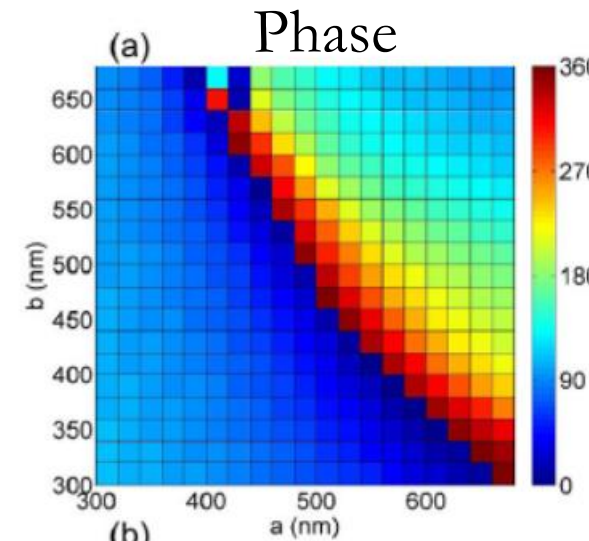
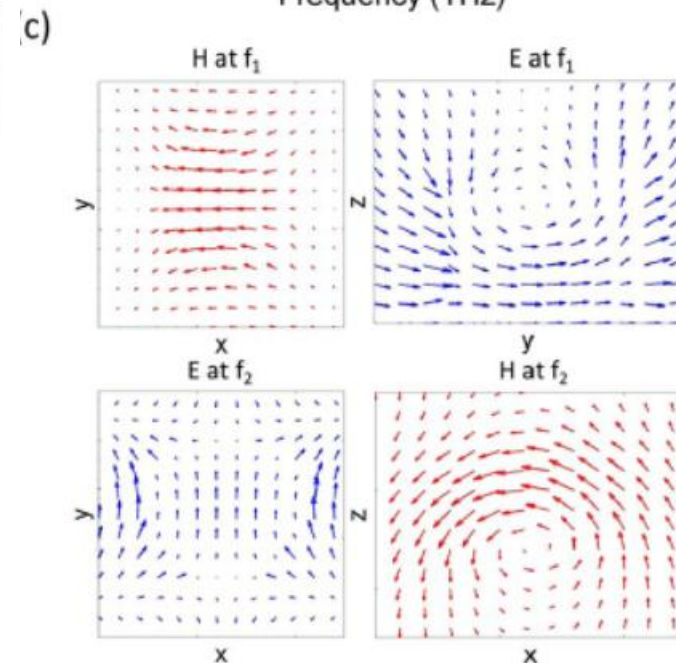
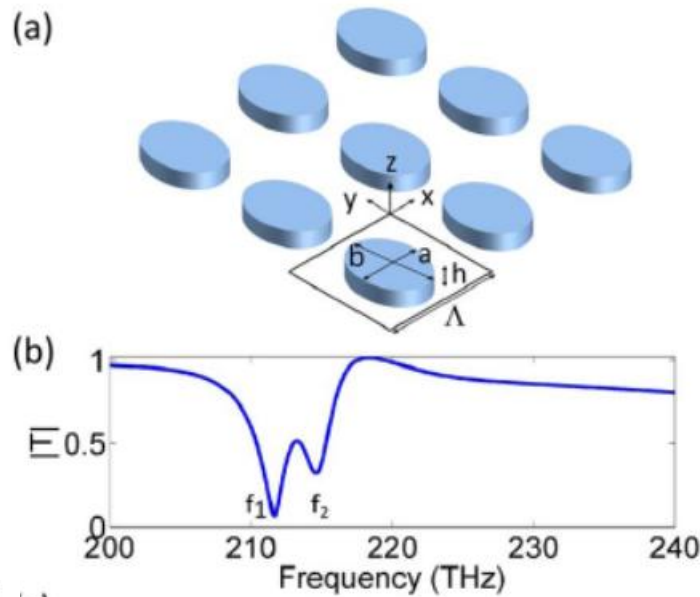


4.2 超表面原理

2. 磁偶极子震荡



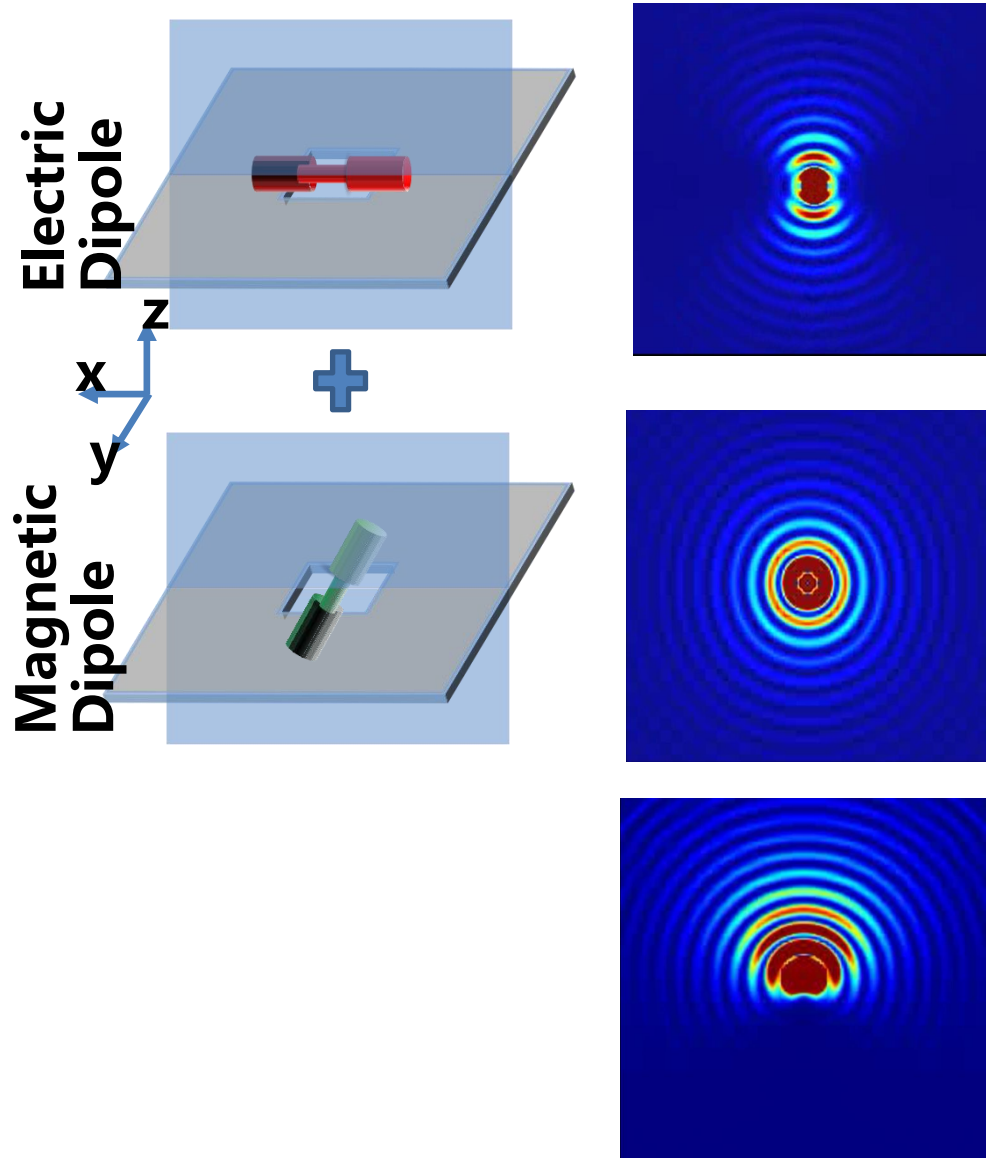
$$\mathbf{B}(\mathbf{x}) = \frac{\mu_0}{4\pi} \left[\frac{3\mathbf{n}(\mathbf{n} \cdot \mathbf{m}) - \mathbf{m}}{|\mathbf{x}|^3} \right]$$



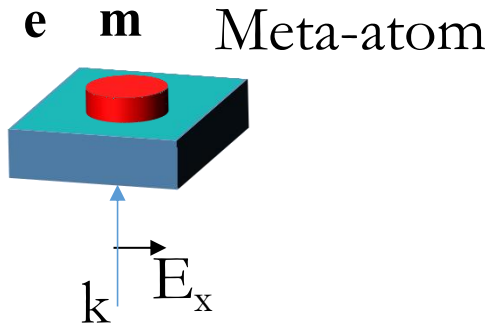
Transmission

4.2 超表面原理

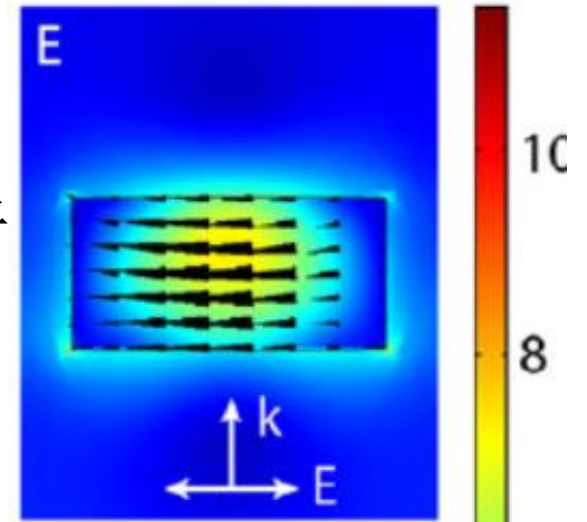
3. 电、磁偶极子震荡



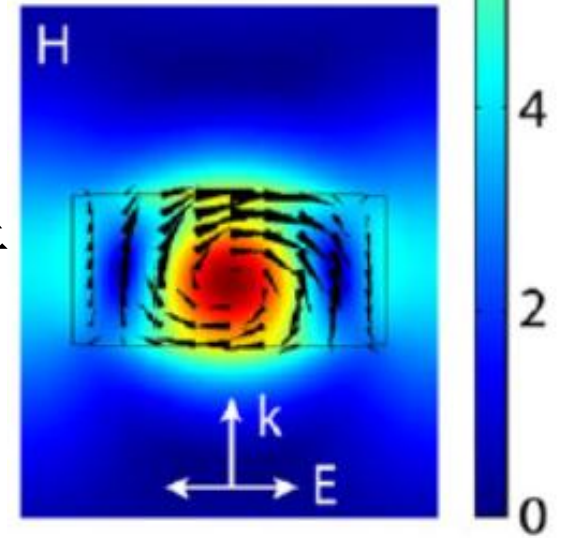
4.2 超表面原理



电偶极子

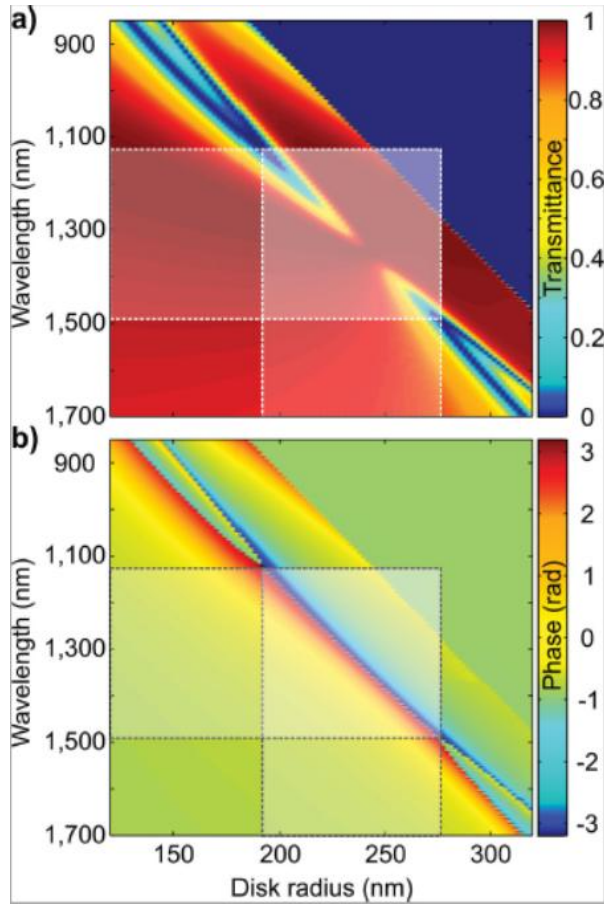


磁偶极子



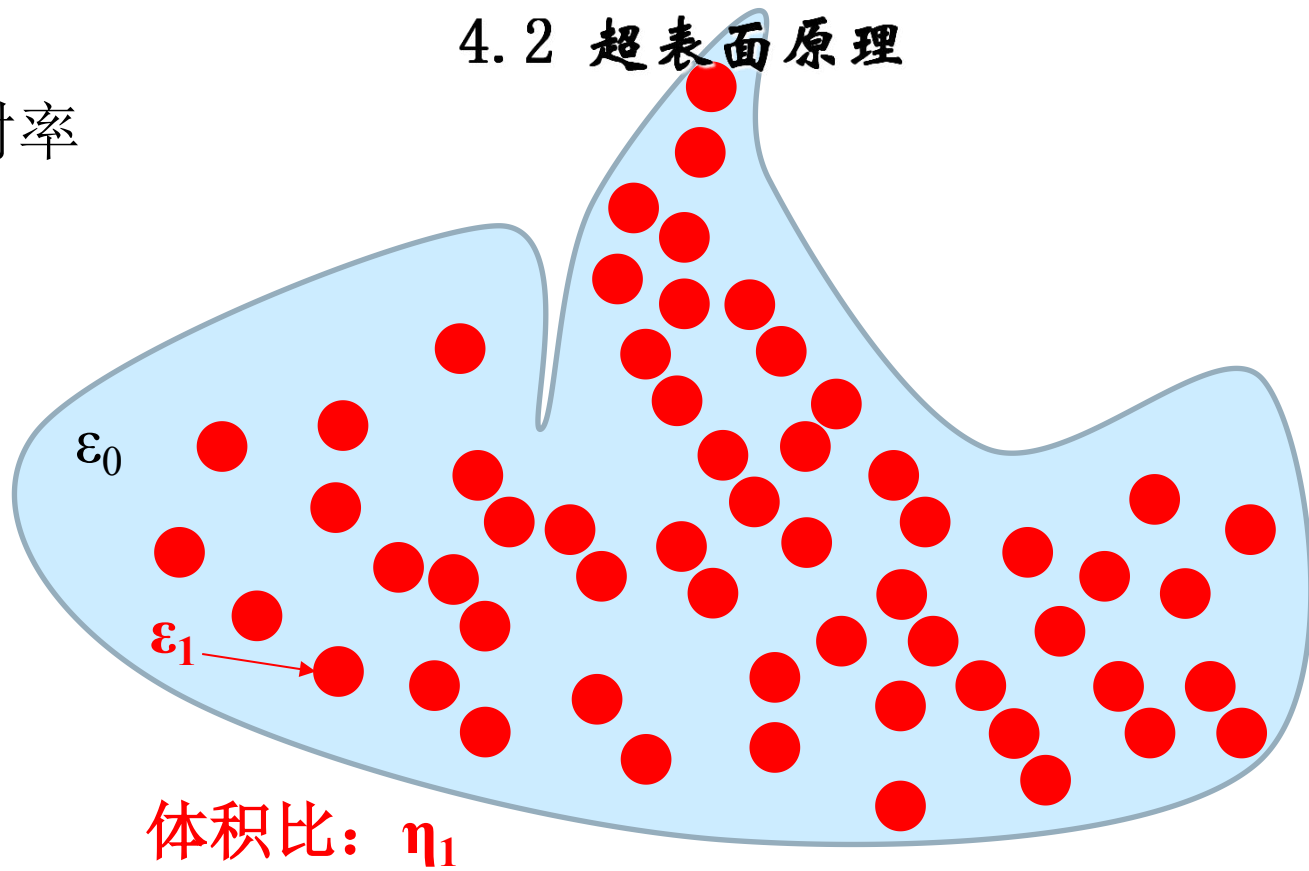
透射率

位相



4.2 超表面原理

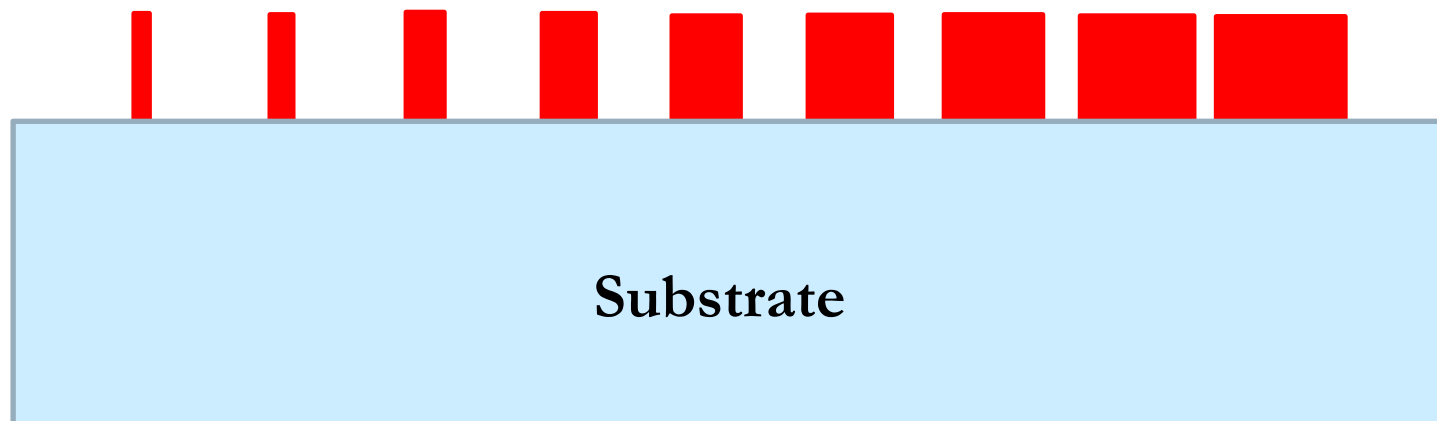
4. 等效折射率



Maxwell-Garnet formula:

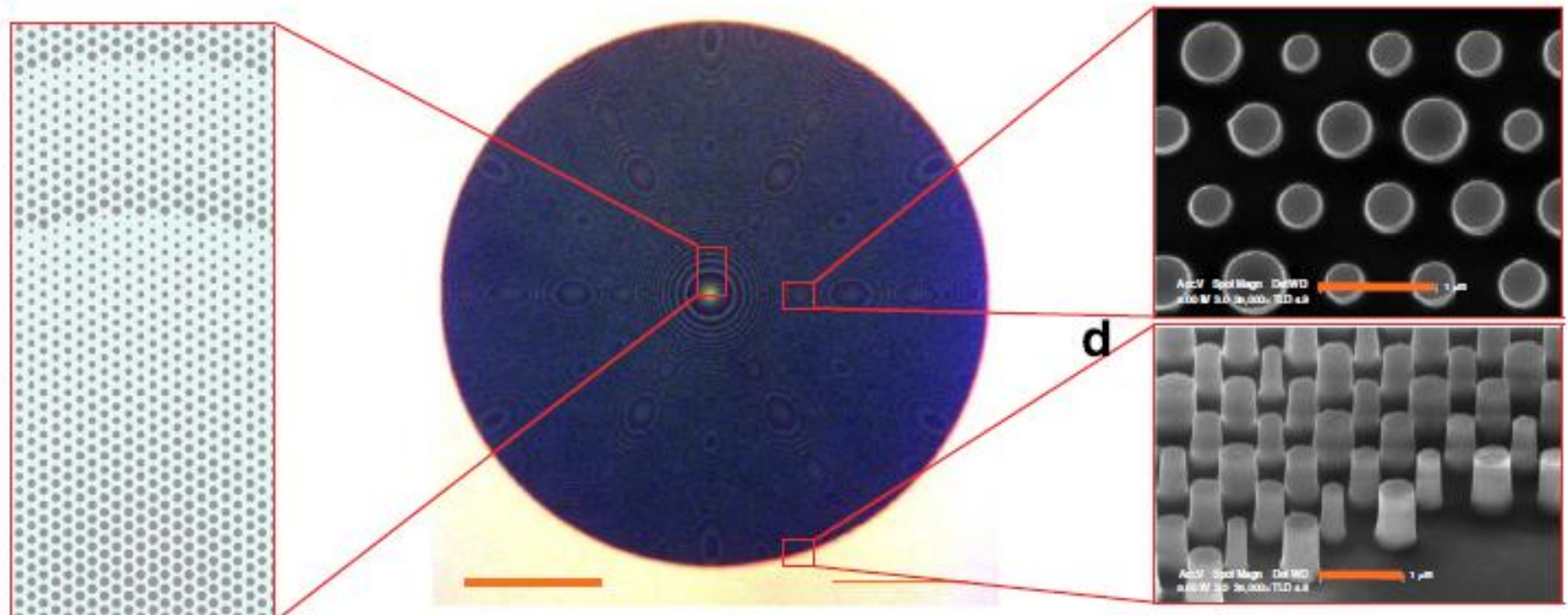
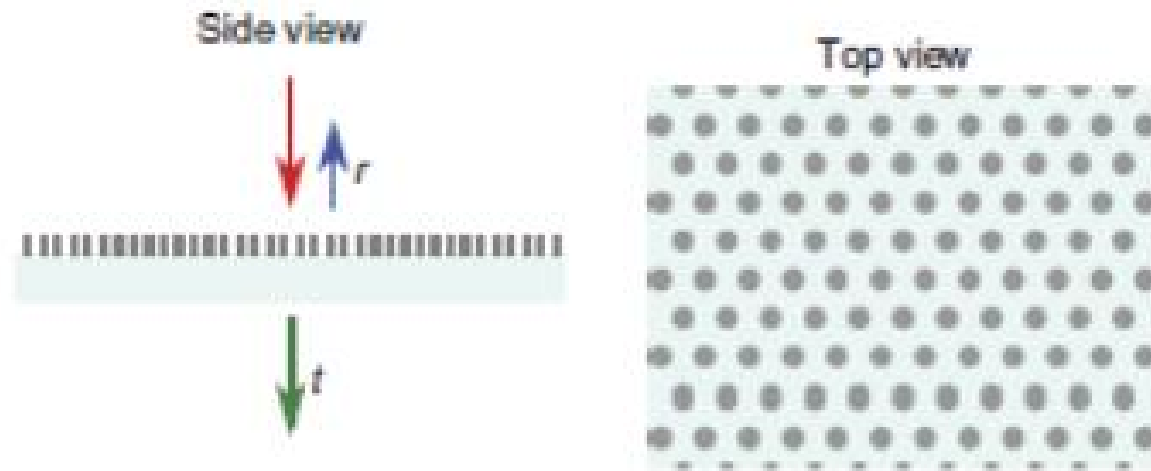
$$\left(\frac{\epsilon - \epsilon_0}{\epsilon + 2\epsilon_0} \right) = \eta_1 \left(\frac{\epsilon_1 - \epsilon_0}{\epsilon_1 + 2\epsilon_0} \right)$$

4.2 超表面原理



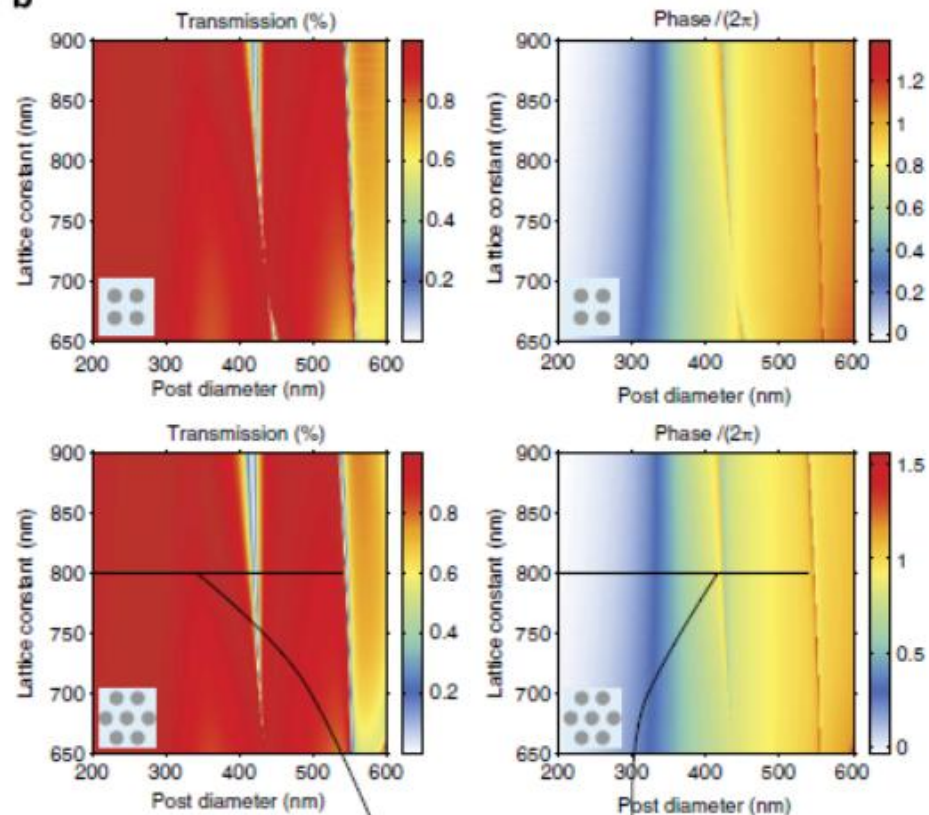
$$\frac{\varepsilon_i - \varepsilon_0}{\varepsilon_i + 2\varepsilon_0} = \eta_i \frac{\varepsilon_1 - \varepsilon_0}{\varepsilon_1 + 2\varepsilon_0}$$

4.2 超表面原理

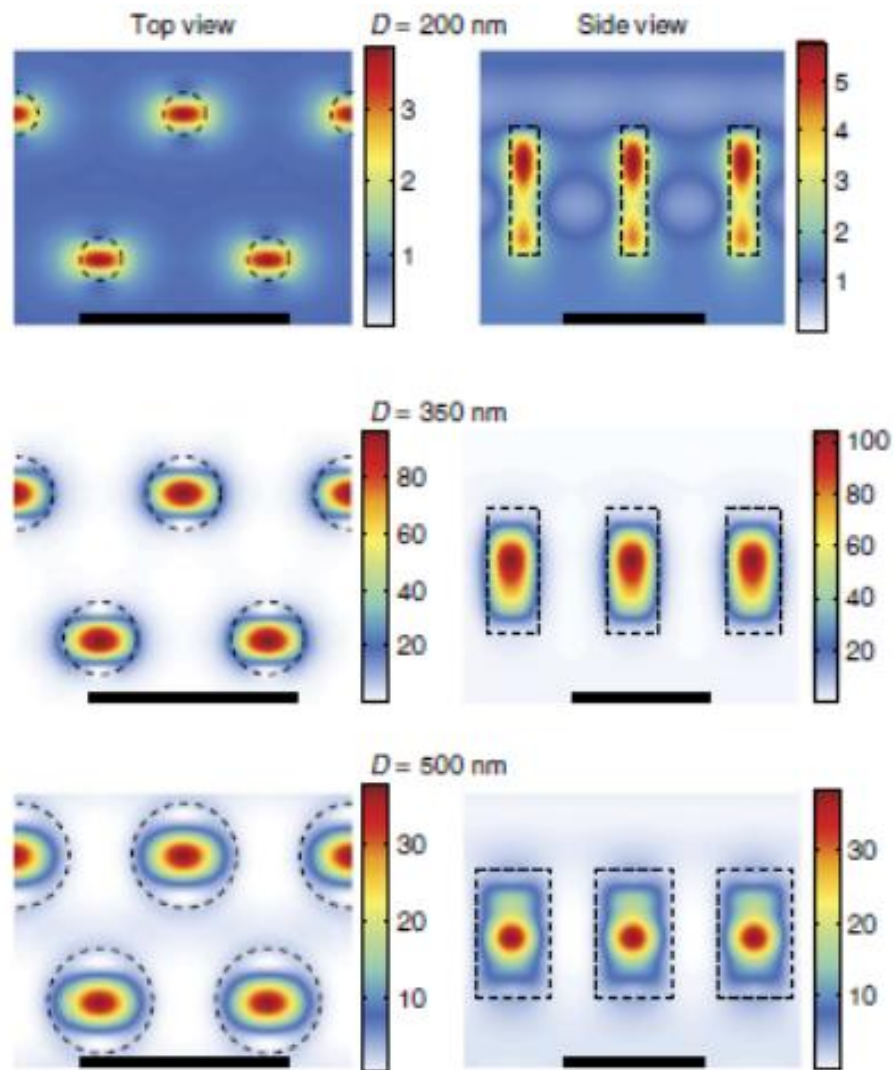
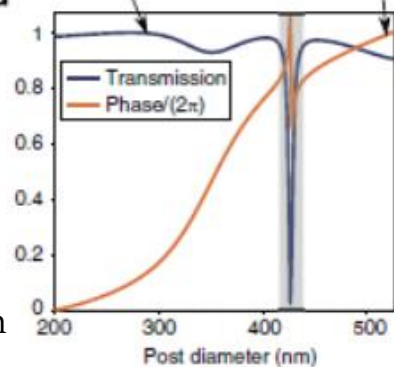


4.2 超表面原理

b



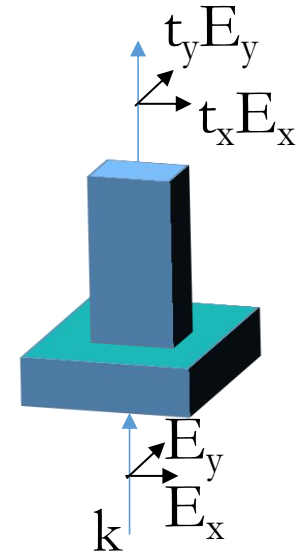
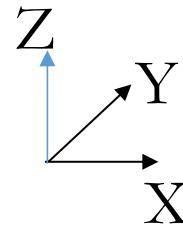
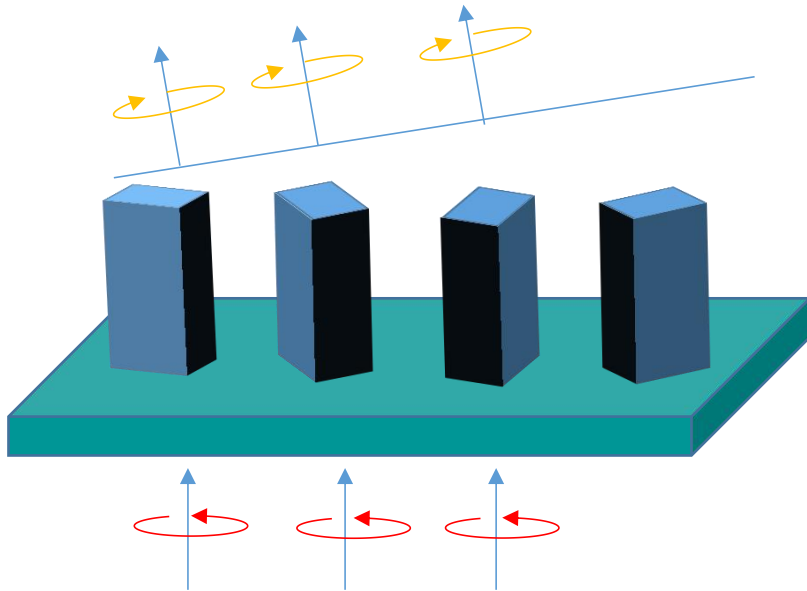
d



Height: 940nm
Period: 800nm
Lambda:1550nm

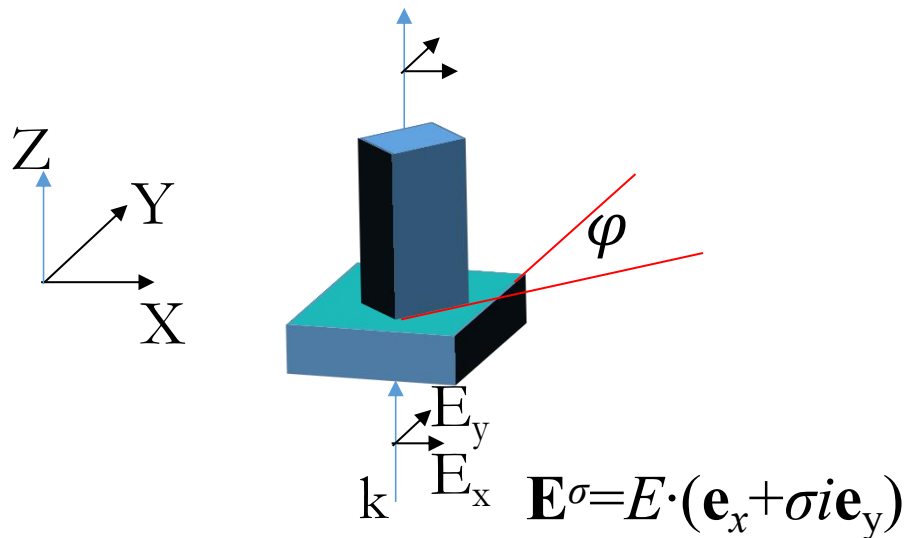
4.2 超表面原理

5. 几何相位型超表面—反铁磁震荡模式



$$\mathbf{T} = \begin{bmatrix} t_x & 0 \\ 0 & t_y \end{bmatrix}$$

几何相位型超表面的工作原理



$$\mathbf{T}(\varphi) = \mathbf{R}(-\varphi) \begin{bmatrix} t_x & 0 \\ 0 & t_y \end{bmatrix} \mathbf{R}(\varphi)$$

$$\mathbf{R}(\varphi) = \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix}$$

$$\begin{aligned} \mathbf{E}_t &= \mathbf{T}(\varphi) \mathbf{E}^\sigma = E \begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} t_x & 0 \\ 0 & t_y \end{bmatrix} \begin{bmatrix} \cos\varphi & \sin\varphi \\ -\sin\varphi & \cos\varphi \end{bmatrix} \begin{bmatrix} 1 \\ \sigma i \end{bmatrix} \\ &= \frac{t_x + t_y}{2} \mathbf{E}^\sigma + \frac{t_x - t_y}{2} e^{2i\sigma\varphi} \mathbf{E}^{-\sigma} \end{aligned}$$

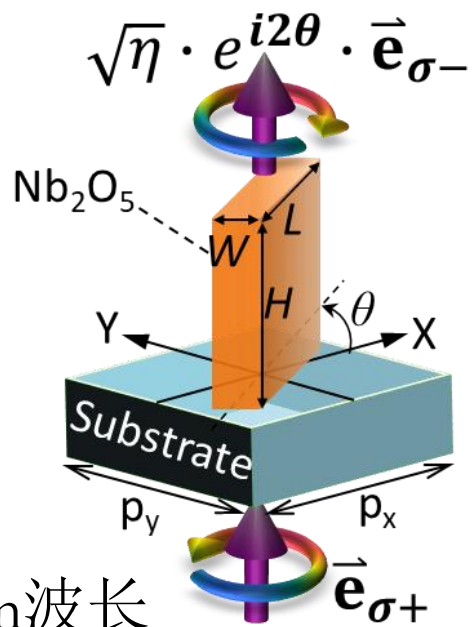
$$\text{当 } t_x = t_y \text{ 时, } \mathbf{E}_t = \frac{t_x + t_y}{2} \mathbf{E}^\sigma$$

无位相调制

$$\text{当 } t_x = -t_y \text{ 时, } \mathbf{E}_t = t_x e^{2i\sigma\varphi} \mathbf{E}^{-\sigma}$$

有几何位相，但要求实现亚波长半波片

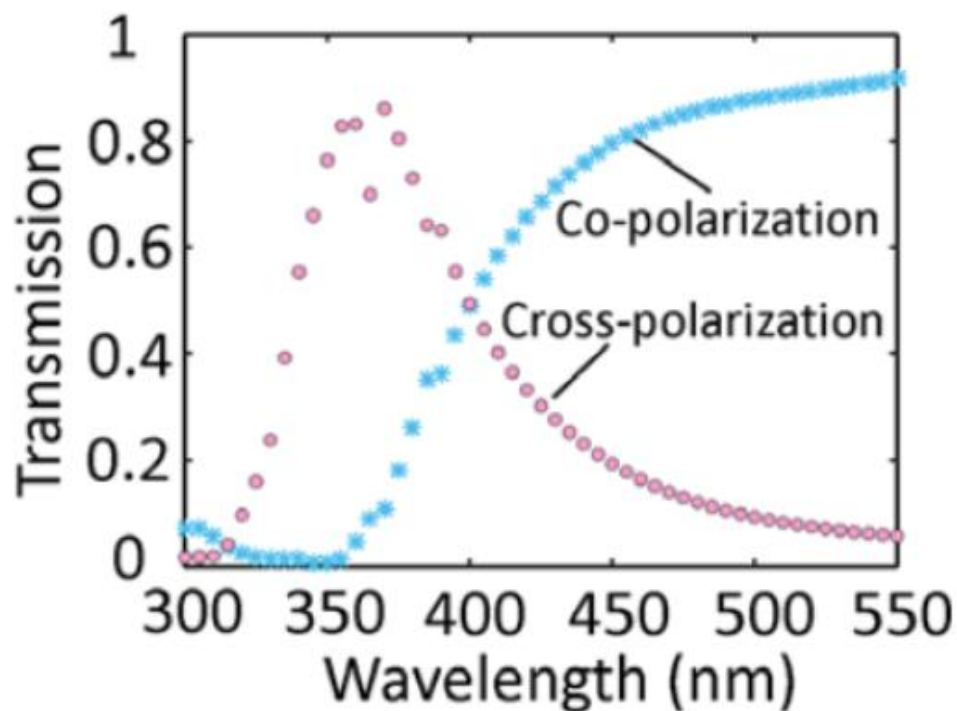
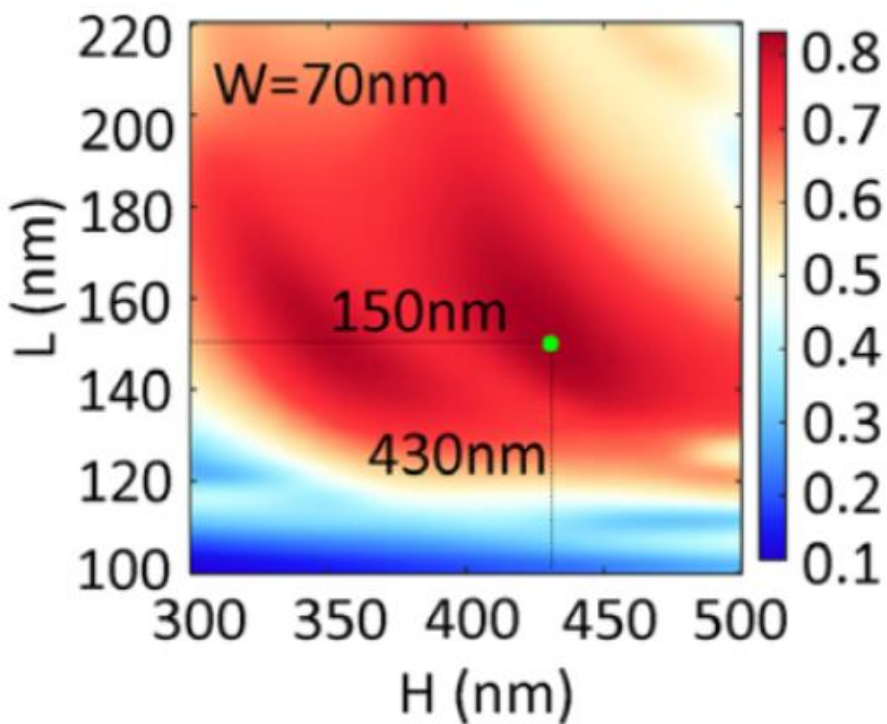
其他情况，透射光中包含Co-polarization and Cross-polarization的光，而Co-polarization光是背景光，需要被滤掉。



通过数值模拟，优化最佳的结构尺寸，使得转换效率最大。

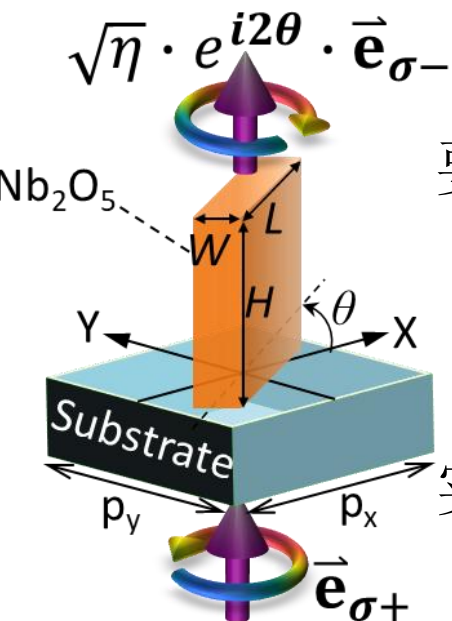
$$\mathbf{E}_{\text{in}}^{\sigma+} = \begin{bmatrix} 1 \\ i \end{bmatrix}$$

$$\mathbf{E}_{\text{out}} = \begin{bmatrix} E_x \\ E_y \end{bmatrix} = \frac{E_x - iE_y}{2} \begin{bmatrix} 1 \\ i \end{bmatrix} + \frac{E_x + iE_y}{2} \begin{bmatrix} 1 \\ -i \end{bmatrix}$$



介质型纳米半波片的宏观和微观机理

i. Macroscopic Approach: Effective medium



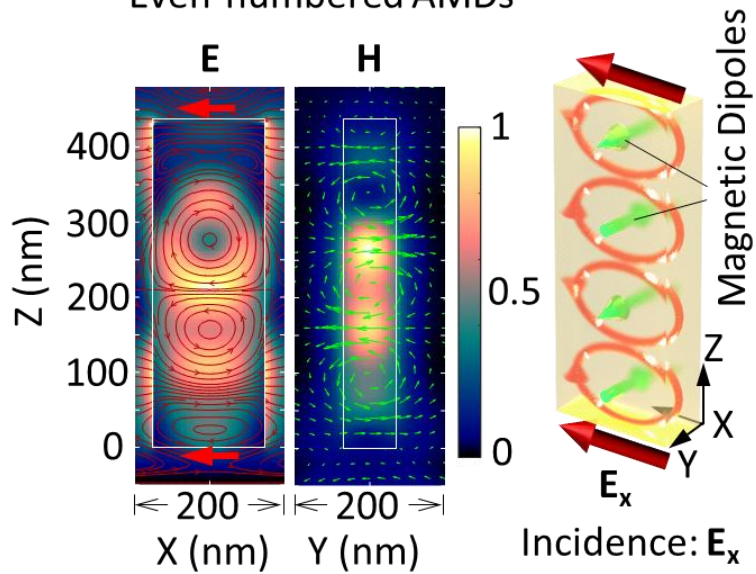
- 要求:
- 1) 对 x 偏振光, 其通过该介质柱后的位相为 ϕ_x
 - 2) 对 y 偏振光, 其通过该介质柱后的位相为 ϕ_y
 - 3) 要实现半波片, 则 $\Delta\phi=\pi$

实际结构:

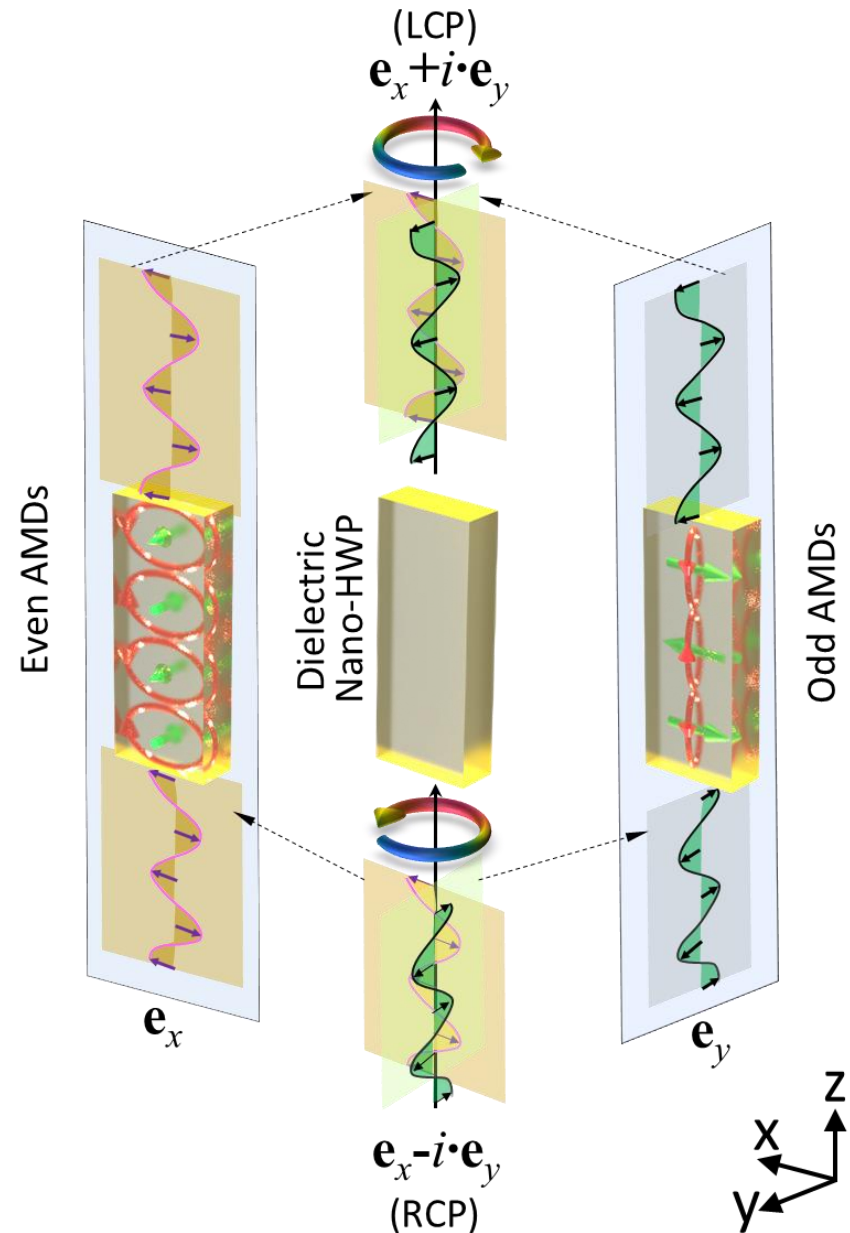
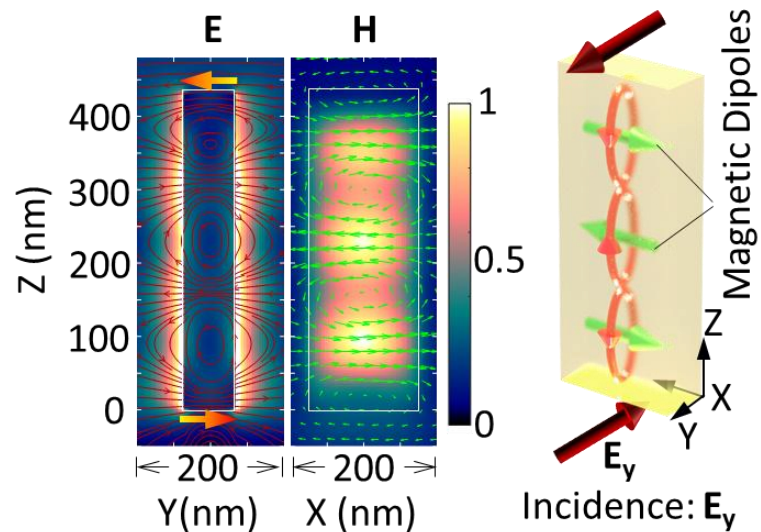
- 1) 结构是平的, 几何光程差为0
- 2) x 方向有效折射率与 y 方向的有效折射率不同, 且差别正好满足 $k(n_x - n_y)h = \pi$

ii. Microscopic Approach: Antiferromagnetic modes

Even-numbered AMDs



Odd-numbered AMDs



工作原理总结：

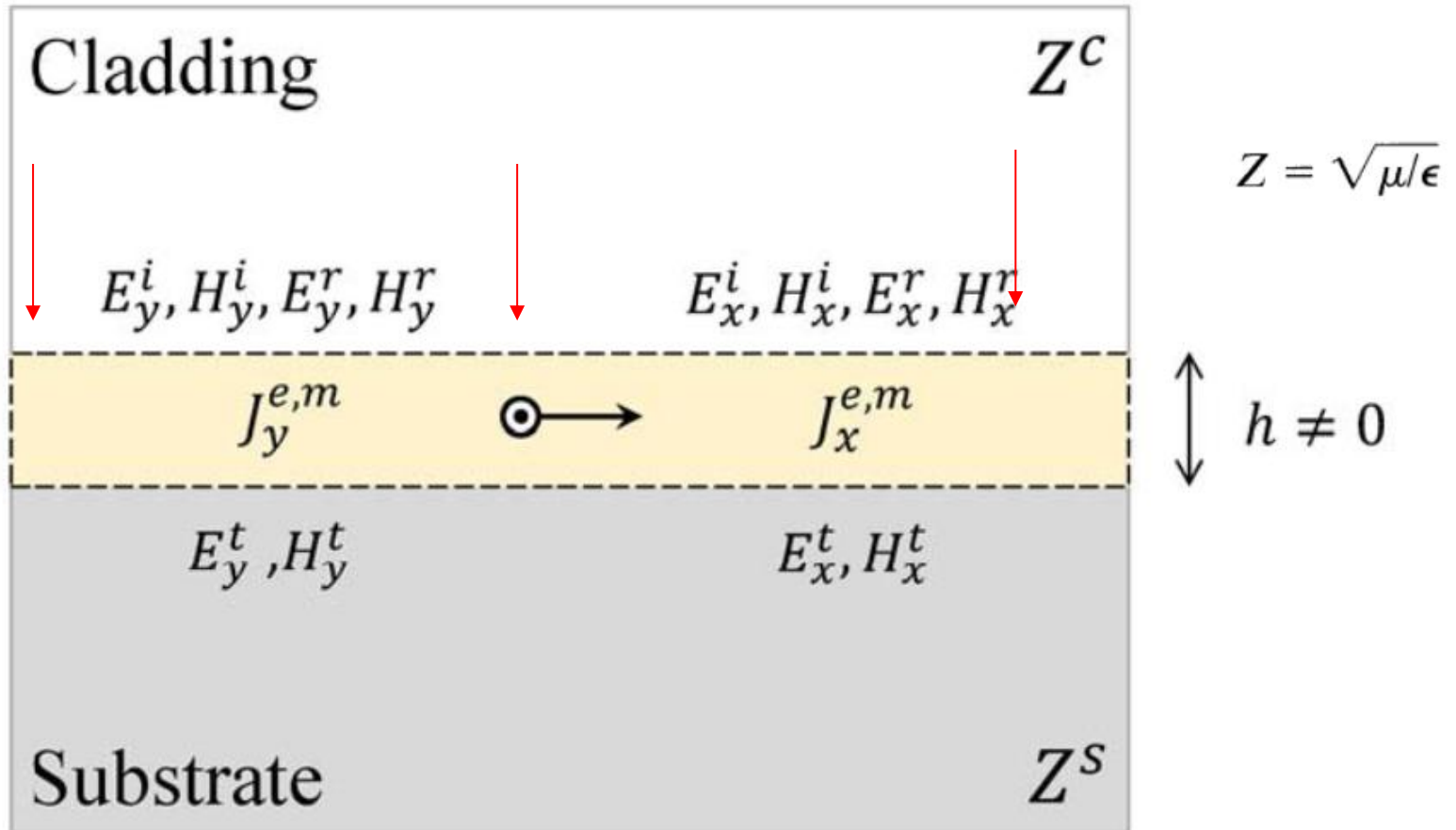
- 1. 物理基础：电、磁偶极子震荡
- 2. 材料：高折射率、低吸收；基底透明、无吸收

波段	透射	反射
红外光	硅	Au
可见光	硅 (>600nm) , TiO ₂ , GaN, Si ₃ N ₄ , GaP	Ag
紫外光	Nb ₂ O ₅ , HfO ₂ , Diamond	Al

- 3. 几何结构：横向尺寸变化（period: $\lambda/2$ ）、纵向尺寸不变(< λ)
- 4. 加工方式：电子束曝光技术

4.3 超表面器件的效率

超表面器件的电动力学模型



$$\bar{J}^e = A^{-1} \int_{-h/2}^{h/2} dz \int_A (\bar{j} + \partial \bar{D} / \partial t) dA$$

$$\bar{J}^m = A^{-1} \int_{-h/2}^{h/2} dz \int_A (\partial \bar{B} / \partial t) dA$$

4.3 超表面器件的效

★

$$\nabla \times \bar{E} = -\frac{\partial \bar{B}}{\partial t}, \quad \nabla \times \bar{H} = \bar{J} + \frac{\partial \bar{D}}{\partial t}$$

边界条件: $(\mathbf{D}_2 - \mathbf{D}_1) \cdot \mathbf{n}_{21} = \sigma \quad (\mathbf{B}_2 - \mathbf{B}_1) \cdot \mathbf{n} = 0$

$(\mathbf{E}_2 - \mathbf{E}_1) \times \mathbf{n}_{21} = 0 \quad \mathbf{n} \times (\mathbf{H}_2 - \mathbf{H}_1) = \mathbf{K}$

$\vec{n} \times (\vec{H}_1 - \vec{H}_2) = \vec{J}^e, \quad \vec{n} \times (\vec{E}_1 - \vec{E}_2) = -\vec{J}^m$

$$\begin{cases} E_x^i + E_x^r - E_x^t = J_y^m \\ E_y^i + E_y^r - E_y^t = -J_x^m \end{cases} \quad \begin{cases} H_x^i = -\frac{1}{Z^c} E_y^i \\ H_y^i = \frac{1}{Z^c} E_x^i \end{cases}, \quad \begin{cases} H_x^r = \frac{1}{Z^c} E_y^r \\ H_y^r = -\frac{1}{Z^c} E_x^r \end{cases}, \quad \begin{cases} H_x^t = -\frac{1}{Z^s} E_y^t \\ H_y^t = \frac{1}{Z^s} E_x^t \end{cases}$$

$$H_x^i + H_x^r - H_x^t = -J_y^e$$

$$H_y^i + H_y^r - H_y^t = J_x^e$$

电极化和磁极化电流密度, 定义:

$$\bar{J}^e \equiv \tilde{\chi}^e \cdot \frac{1}{Z^c} \bar{E}^i$$

$$\bar{J}^m \equiv \tilde{\chi}^m \cdot Z^c \bar{H}^i = -\tilde{\chi}^m \cdot i\hat{\sigma}_3 \cdot \bar{E}^i$$

$$\hat{\sigma}_3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

4.3 超表面器件的效率

$$\mathbf{T} = \frac{2}{1+Z^c/Z^s} \hat{I} - \frac{1}{1+Z^c/Z^s} \bar{\chi}^e - \frac{1}{1+Z^c/Z^s} \hat{\sigma}_3 \cdot \bar{\chi}^m \cdot \hat{\sigma}_3$$

$$\mathbf{R} = \frac{1-Z^c/Z^s}{1+Z^c/Z^s} \hat{I} - \frac{1}{1+Z^c/Z^s} \bar{\chi}^e + \frac{Z^c/Z^s}{1+Z^c/Z^s} \hat{\sigma}_3 \cdot \bar{\chi}^m \cdot \hat{\sigma}_3$$

$$\hat{I} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\hat{\sigma}_3 = \begin{pmatrix} 1 & \\ & -1 \end{pmatrix}$$

$$Z^c = Z^s = Z^0$$

镜向对称

$$\mathbf{T} = \hat{I} - \frac{1}{2} \bar{\chi}^e - \frac{1}{2} \hat{\sigma}_3 \cdot \bar{\chi}^m \cdot \hat{\sigma}_3 \quad \bar{\chi}^e = \begin{pmatrix} \chi_u^e & \\ & \chi_v^e \end{pmatrix} \quad \mathbf{T} = \begin{bmatrix} t_{uu} & 0 \\ 0 & t_{vv} \end{bmatrix}$$

$$\mathbf{R} = -\frac{1}{2} \bar{\chi}^e + \frac{1}{2} \hat{\sigma}_3 \cdot \bar{\chi}^m \cdot \hat{\sigma}_3 \quad \bar{\chi}^m = \begin{pmatrix} \chi_u^m & \\ & \chi_v^m \end{pmatrix} \quad \mathbf{R} = \begin{bmatrix} r_{uu} & 0 \\ 0 & r_{vv} \end{bmatrix}$$

$$t_{uu} = 1 - \frac{1}{2} \chi_u^e - \frac{1}{2} \chi_v^m, \quad t_{vv} = 1 - \frac{1}{2} \chi_v^e - \frac{1}{2} \chi_u^m,$$

$$r_{uu} = -\frac{1}{2} \chi_u^e + \frac{1}{2} \chi_v^m, \quad r_{vv} = -\frac{1}{2} \chi_v^e + \frac{1}{2} \chi_u^m.$$

4.3 超表面器件的效

率

$$\mathbf{T} = \begin{bmatrix} t_{uu} & 0 \\ 0 & t_{vv} \end{bmatrix} = \frac{t_{uu} + t_{vv}}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{t_{uu} - t_{vv}}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\mathbf{R} = \begin{bmatrix} r_{uu} & 0 \\ 0 & r_{vv} \end{bmatrix} = \frac{r_{uu} + r_{vv}}{2} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \frac{r_{uu} - r_{vv}}{2} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{aligned} T_n &\equiv |(t_{uu} + t_{vv})|^2/4, & R_n &\equiv |(r_{uu} + r_{vv})|^2/4, & \text{Co-polarization: } & \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \\ T_a &\equiv |(t_{uu} - t_{vv})|^2/4, & R_a &\equiv |(r_{uu} - r_{vv})|^2/4, & \text{Cross-polarization: } & \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \end{aligned}$$

$$t_{uu} = 1 - \frac{1}{2}\chi_u^e - \frac{1}{2}\chi_v^m, \quad t_{vv} = 1 - \frac{1}{2}\chi_v^e - \frac{1}{2}\chi_u^m,$$

$$r_{uu} = -\frac{1}{2}\chi_u^e + \frac{1}{2}\chi_v^m, \quad r_{vv} = -\frac{1}{2}\chi_v^e + \frac{1}{2}\chi_u^m.$$

$$T_n = \left| 1 - \frac{1}{4}(\chi_u^e + \chi_v^m) - \frac{1}{4}(\chi_v^e + \chi_u^m) \right|^2$$

$$T_a = \left| -\frac{1}{4}(\chi_u^e + \chi_v^m) + \frac{1}{4}(\chi_v^e + \chi_u^m) \right|^2$$

$$R_n = \left| \frac{1}{4}(\chi_u^e - \chi_v^m) + \frac{1}{4}(\chi_v^e - \chi_u^m) \right|^2$$

$$R_a = \left| \frac{1}{4}(\chi_u^e - \chi_v^m) - \frac{1}{4}(\chi_v^e - \chi_u^m) \right|^2$$

4.3 超表面器件的效率

对于透射型器件，要实现高转换效率，则要求

$$T_n=0, R_n=0, R_a=0, T_a=1$$

$$T_n = \left| 1 - \frac{1}{4}(\chi_u^e + \chi_v^m) - \frac{1}{4}(\chi_v^e + \chi_u^m) \right|^2$$

$$T_a = \left| -\frac{1}{4}(\chi_u^e + \chi_v^m) + \frac{1}{4}(\chi_v^e + \chi_u^m) \right|^2$$

$$R_n = \left| \frac{1}{4}(\chi_u^e - \chi_v^m) + \frac{1}{4}(\chi_v^e - \chi_u^m) \right|^2$$

$$R_a = \left| \frac{1}{4}(\chi_u^e - \chi_v^m) - \frac{1}{4}(\chi_v^e - \chi_u^m) \right|^2$$

$$\chi_u^e = \chi_v^m > 0 \quad \chi_v^e = \chi_u^m > 0$$



$$T_n = \left| 1 - \frac{1}{2}\chi_u^e - \frac{1}{2}\chi_v^e \right|^2$$

$$T_a = \left| -\frac{1}{2}\chi_u^e + \frac{1}{2}\chi_v^e \right|^2$$

特解： $\chi_u^e = \chi_v^m = 1 + e^{i\delta}$
 $\chi_v^e = \chi_u^m = 1 - e^{i\delta}$

该解指出，要想实现100%效率转换，超表面内部必须有电和磁震荡即，超表面器件的效率极限是100%

4.3 超表面器件的效率

率

只有电震荡或磁震荡的超表面，其效率极限为25%!!!

$$T_n = \left| 1 - \frac{1}{4}(\chi_u^e + \chi_v^m) - \frac{1}{4}(\chi_v^e + \chi_u^m) \right|^2$$

$$T_a = \left| -\frac{1}{4}(\chi_u^e + \chi_v^m) + \frac{1}{4}(\chi_v^e + \chi_u^m) \right|^2$$

$$R_n = \left| \frac{1}{4}(\chi_u^e - \chi_v^m) + \frac{1}{4}(\chi_v^e - \chi_u^m) \right|^2$$

$$R_a = \left| \frac{1}{4}(\chi_u^e - \chi_v^m) - \frac{1}{4}(\chi_v^e - \chi_u^m) \right|^2$$



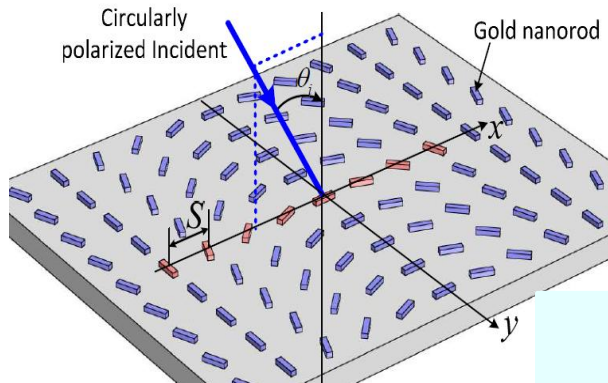
$$T_n = \left| 1 - \frac{1}{4}(\chi_u^e + \chi_v^e) \right|^2, \quad T_a = \left| \frac{1}{4}(\chi_u^e - \chi_v^e) \right|^2$$

$$R_n = \left| \frac{1}{4}(\chi_u^e + \chi_v^e) \right|^2, \quad R_a = \left| \frac{1}{4}(\chi_u^e - \chi_v^e) \right|^2$$

$$(\chi_u^e + \chi_v^e)/4 = x + iy$$

$$T_n + R_n = 2(x - 1/2)^2 + 2y^2 + \frac{1}{2} \geq \frac{1}{2}$$

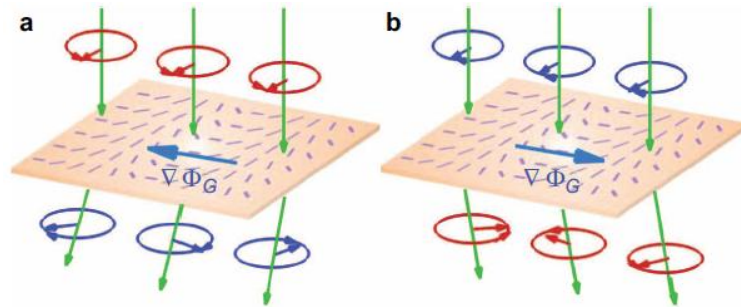
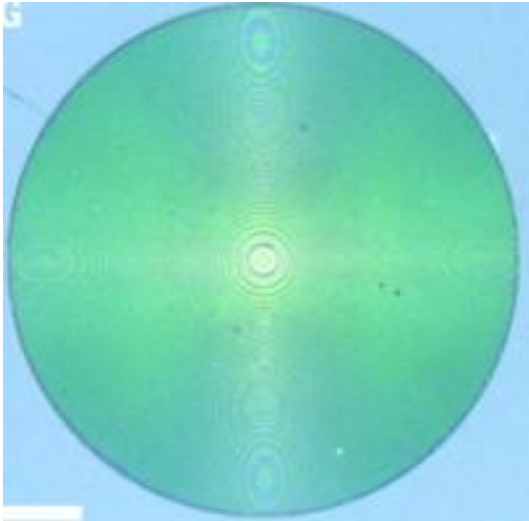
$$T_a + R_a + T_n + R_n = 1$$



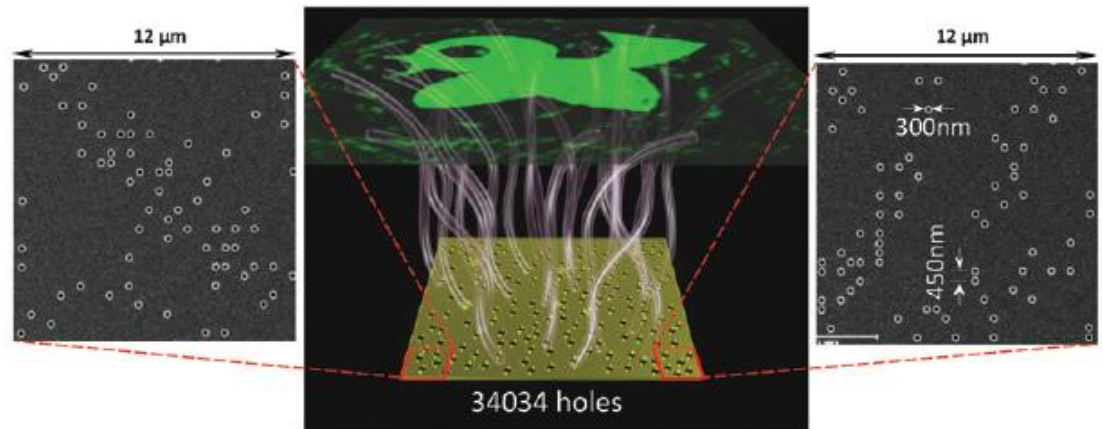
4.4 超表面的应用

光栅

透镜

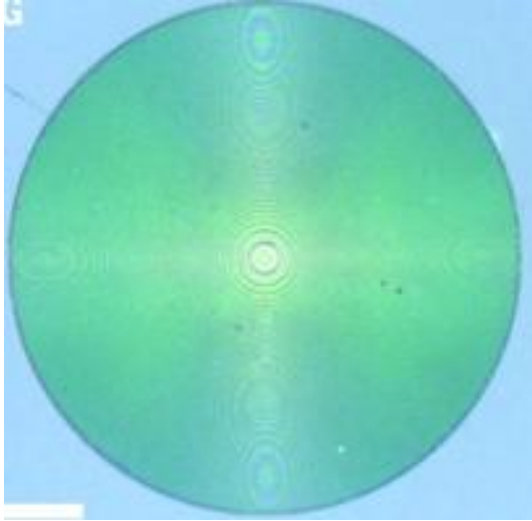


全息



4.4 超表面的应用

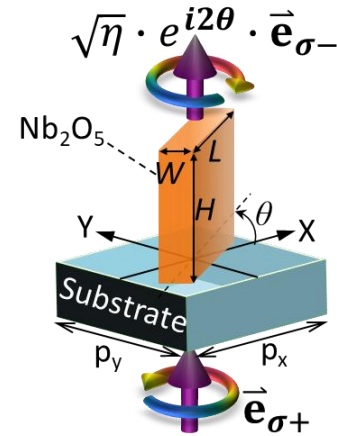
透镜



$$E = e^{i\varphi}$$

$$\varphi(r) = \frac{2\pi}{\lambda} \left(f - \sqrt{f^2 + r^2} \right)$$

几何位相



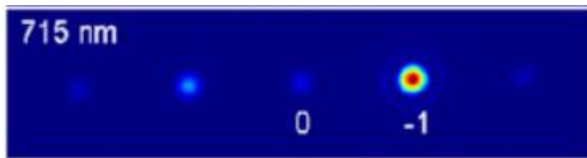
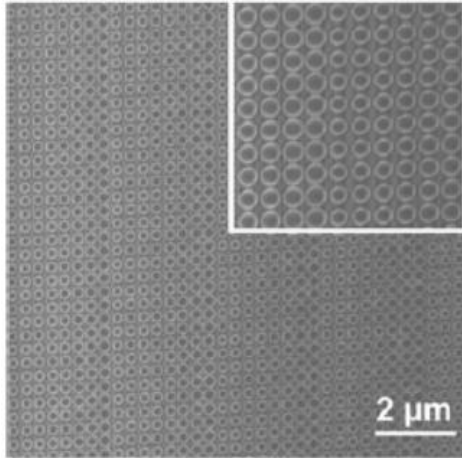
$$E = e^{i\sigma\varphi}$$

当 $\sigma = 1$ ，凸透镜位相，光会聚

当 $\sigma = -1$ ，凹透镜位相，光发散

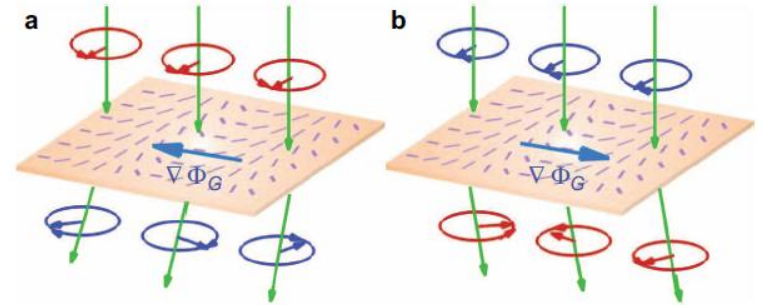
4.4 超表面的应用

闪耀光栅

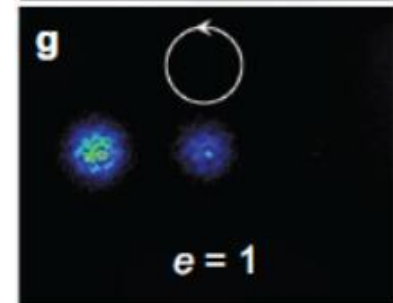
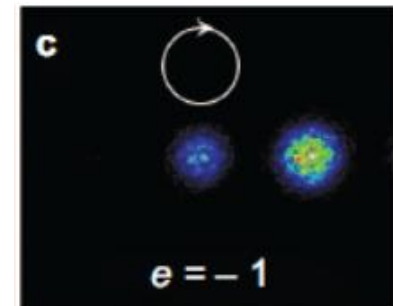


$$E = e^{ik\sin\theta x}$$

几何位相型闪耀光栅

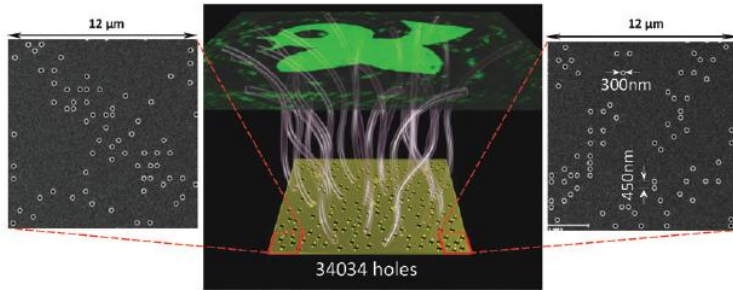


$$E = e^{i\sigma k\sin\theta x}$$

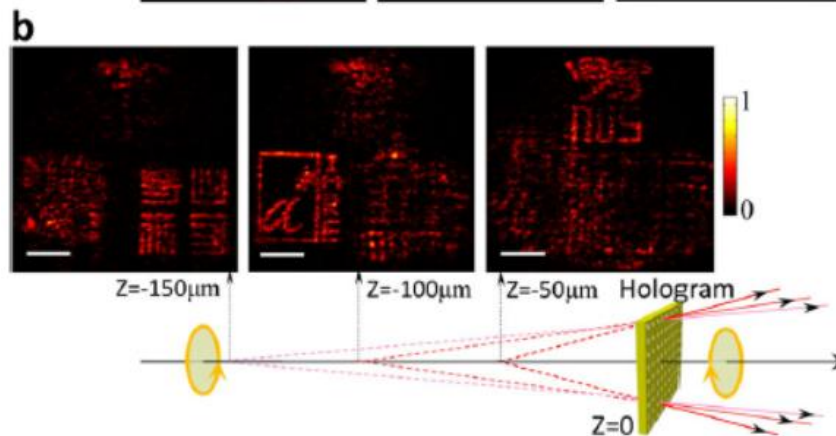
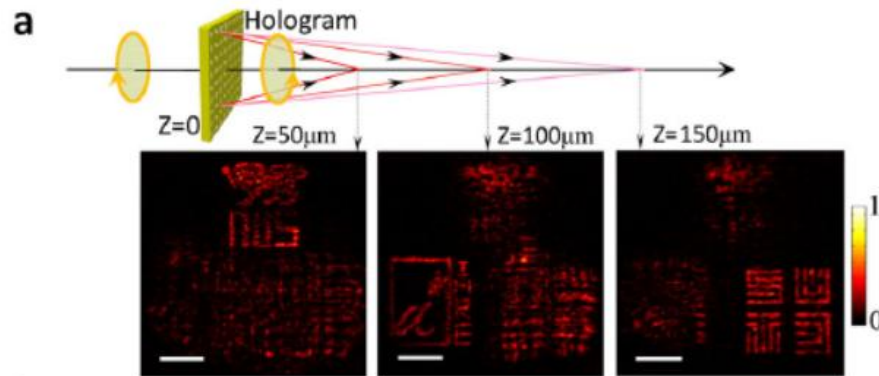
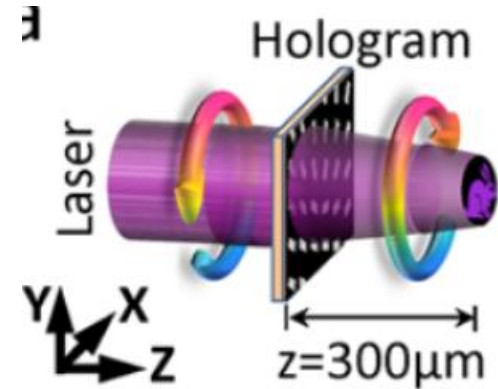


4.4 超表面的应用

结构变化型全息



几何位相型全息



4.4 超表面的应用

几何位相型Fresnel全息

$$E_{\sigma}(x, y, z') = \frac{\exp(ikz)}{i\lambda z} \iint A(x_0, y_0) e^{i\phi(x_0, y_0)} \exp\left\{i \frac{k}{2z'} [(x - x_0)^2 + (y - y_0)^2]\right\} dx_0 dy_0$$

$$E_{-\sigma}(x, y, z') = \frac{\exp(ikz')}{i\lambda z'} \iint A(x_0, y_0) e^{-i\phi(x_0, y_0)} \exp\left\{i \frac{k}{2z'} [(x - x_0)^2 + (y - y_0)^2]\right\} dx_0 dy_0$$

$$I_{-\sigma}(x, y, z) = |E_{-\sigma}(x, y, z')|^2 = |E_{-\sigma}(x, y, z')^*|^2$$

$$= \left| \frac{\exp(-ikz')}{i\lambda(-z')} \iint A^*(x_0, y_0) e^{i\phi(x_0, y_0)} \exp\left\{i \frac{k}{2(-z')} [(x - x_0)^2 + (y - y_0)^2]\right\} dx_0 dy_0 \right|^2$$

$$= |E_{\sigma}(x, y, -z')|^2 = I_{\sigma}(x, y, -z')$$

4.4 超表面的应用

几何位相型Fraunhofer全息

$$\begin{aligned} I_{\sigma}(x, y, z') &= \frac{1}{(\lambda z')^2} \left| \iint A(x_0, y_0) e^{i\phi(x_0, y_0)} \exp\left[-i \frac{k}{z'} (xx_0 + yy_0)\right] dx_0 dy_0 \right|^2 \\ I_{-\sigma}(x, y, z') &= \frac{1}{(\lambda z')^2} \left| \iint A(x_0, y_0) e^{-i\phi(x_0, y_0)} \exp\left[-i \frac{k}{z'} (xx_0 + yy_0)\right] dx_0 dy_0 \right|^2 \\ &= \frac{1}{(\lambda z')^2} \left| \iint A^*(x_0, y_0) e^{i\phi(x_0, y_0)} \exp\left[-i \frac{k}{z'} (-xx_0 - yy_0)\right] dx_0 dy_0 \right|^2 = I_{\sigma}(-x, -y, z') \\ &= \frac{1}{(\lambda z')^2} \left| \iint A^*(x_0, y_0) e^{i\phi(x_0, y_0)} \exp\left[-i \frac{k}{-z'} (xx_0 + yy_0)\right] dx_0 dy_0 \right|^2 = I_{\sigma}(x, y, -z') \end{aligned}$$

4.4 超表面的应用

<https://1252412222.vod2.myqcloud.com/794b4a11vodbj1252412222/ebfc42385285890800749988324/f0.mp4>

<https://1252412222.vod2.myqcloud.com/794b4a11vodbj1252412222/7ac59eb25285890800799314474/f0.mp4>

4.5 超表面器件性能的电磁计算模拟

——时域有限差分法 (FDTD)

1. 麦克斯韦方程的基础知识
2. 一维和三维Maxwell方程的Yee算法
3. 数值稳定性分析
4. 吸收边界条件
5. 波源的设置

基础知识

麦克斯韦方程微分形式：

$$\nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\nabla \cdot \mathbf{D} = \rho$$

BPM
方式

Maxwell Equations:

$$\begin{cases} \nabla \times \vec{E} = -j\omega\mu_0\vec{H} \\ \nabla \times \vec{H} = j\omega\varepsilon_0\varepsilon_r\vec{E} \end{cases}$$

对于该波导的 TE 模，有 $E_x = E_z = H_y = 0$ ，另外，任何场对 y 方向的导数皆为 0，

因此，Maxwell Equations 可以得到化简，最后得到一组方程： $E_y(x, y, z) = \phi(x, y, |z) \exp(-j\beta z)$

$$\begin{cases} \frac{\partial^2 E_y}{\partial x^2} + \frac{\partial^2 E_y}{\partial z^2} + k_0^2 \varepsilon_r E_y = 0 \\ \frac{\partial^2 H_x}{\partial x^2} + \frac{\partial^2 H_x}{\partial z^2} + k_0^2 \varepsilon_r H_x = 0 \end{cases}$$

$$2j\beta \frac{\partial \phi}{\partial z} - \frac{\partial^2 \phi}{\partial z^2} = \frac{\partial^2 \phi}{\partial x^2} + k_0^2 (\varepsilon_r - n_{\text{eff}}^2) \phi$$

基础知识

麦克斯韦方程微分形式：

FDTD方式将时间进行差分，并且磁场与电场交替迭代更新

对于有耗媒质：

$$\nabla \times H = \frac{\partial D}{\partial t} + J_e, J_e = \sigma E$$

$$\nabla \times E = -\frac{\partial B}{\partial t} - J_m, J_m = sH$$

$$\nabla \cdot B = \rho_m$$

$$\nabla \cdot D = \rho_e$$

$$\nabla \times H = \frac{\partial D}{\partial t} + J$$

$$\nabla \times E = -\frac{\partial B}{\partial t}$$

$$\nabla \cdot B = 0$$

$$\nabla \cdot D = \rho$$

时谐场形式：

$$\nabla \times H = j\omega(\varepsilon - j\frac{\sigma}{\omega})E$$

$$\nabla \times E = -j\omega\mu H$$

$$E(x, y, z, t) = E_0(x, y, z, t)e^{j\omega t}$$

$$H(x, y, z, t) = H_0(x, y, z, t)e^{j\omega t}$$

一维Maxwell方程的Yee算法

一维Maxwell方程，介质参数和场量均与x，y无关(无损，电导率和磁导率为0)

$$\frac{\partial E_x}{\partial t} = -\frac{1}{\epsilon_0 \epsilon_r} \frac{\partial H_y}{\partial z} \qquad \frac{\partial H_y}{\partial t} = -\frac{1}{\mu_0 \mu_r} \frac{\partial E_x}{\partial z}$$

利用一阶导数的二阶中心差分近似，上面的方程变为

$$\frac{E_x^{n+1}(k) - E_x^n(k)}{\Delta t} = -\frac{1}{\epsilon_0 \epsilon_r(k)} \frac{H_y^{n+1/2}(k + \frac{1}{2}) - H_y^{n+1/2}(k - \frac{1}{2})}{\Delta z}$$

$$\frac{H_y^{n+1/2}(k + \frac{1}{2}) - H_y^{n-1/2}(k + \frac{1}{2})}{\Delta t} = -\frac{1}{\mu_0 \mu_r(k + \frac{1}{2})} \frac{E_x^n(k + 1) - E_x^n(k)}{\Delta z}$$

一维Maxwell方程的Yee算法

采用归一化磁场 $\tilde{H} = \sqrt{\frac{\mu_0}{\epsilon_0}} H$

使得电场与归一化磁场有相同的数量级，于是可以得到FDTD迭代公式为

$$\begin{aligned}\tilde{H}_y^{n+1/2}(k + \frac{1}{2}) &= \tilde{H}_y^{n-1/2}(k + \frac{1}{2}) - \frac{c\Delta t}{\Delta z \mu_r(k + \frac{1}{2})} [E_x^n(k+1) - E_x^n(k)] \\ E_x^{n+1}(k) &= E_x^n(k) - \frac{c\Delta t}{\Delta z \epsilon_r(k)} \left[\tilde{H}_y^{n+1/2}(k + \frac{1}{2}) - \tilde{H}_y^{n+1/2}(k - \frac{1}{2}) \right]\end{aligned}$$

式中， $c = \frac{1}{\sqrt{\mu_0 \epsilon_0}}$ 为自由空间中的光速。

一维Maxwell方程的Yee算法

用计算机语言表示的FDTD公式

$$Hy[k] = Hy[k] - ca[k] * (Ex[k + 1] - Ex[k])$$

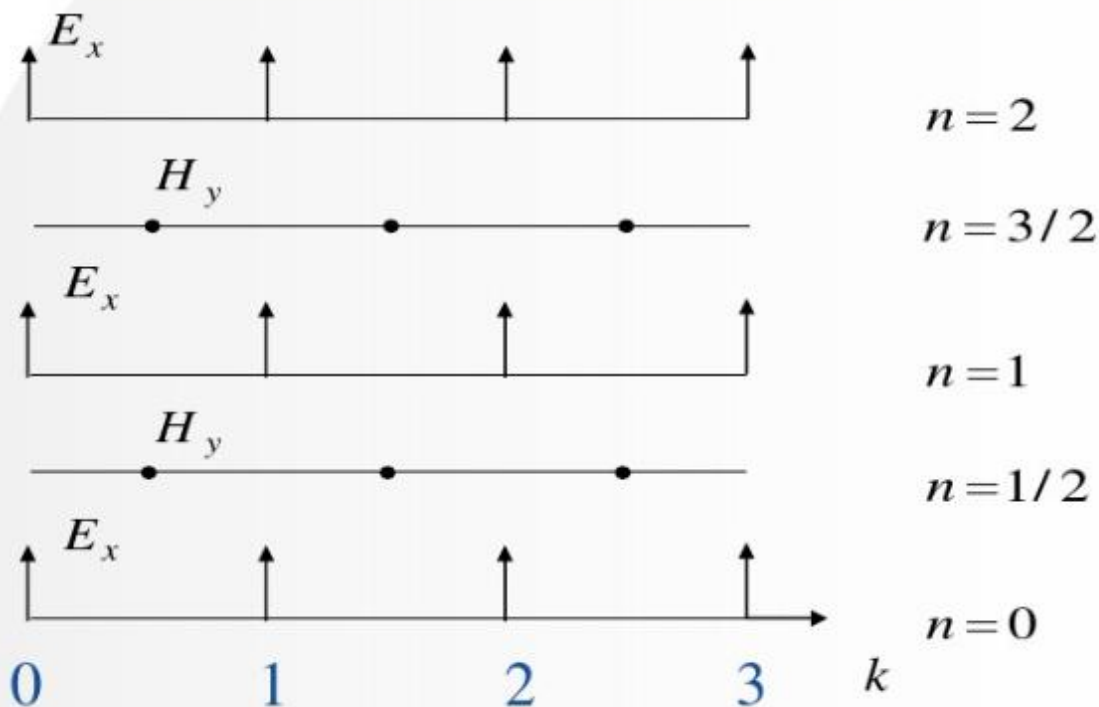
$$Ex[k] = Ex[k] - cb[k] * (Hy[k] - Hy[k - 1])$$

式中，时间变量已隐含在迭代公式中，以及

$$Hy[k] = \tilde{H}_y(k + \frac{1}{2}); Ex[k] = E_x[k];$$
$$ca[k] = \frac{c\Delta t}{\Delta z\mu_r(k + \frac{1}{2})}; cb[k] = \frac{c\Delta t}{\Delta z\varepsilon_r(k)}$$

只要给定了所有空间点上电/磁场的初值，就可以一步一步地求出任意时刻所有空间点上的电/磁场值。

一维Maxwell方程的Yee算法

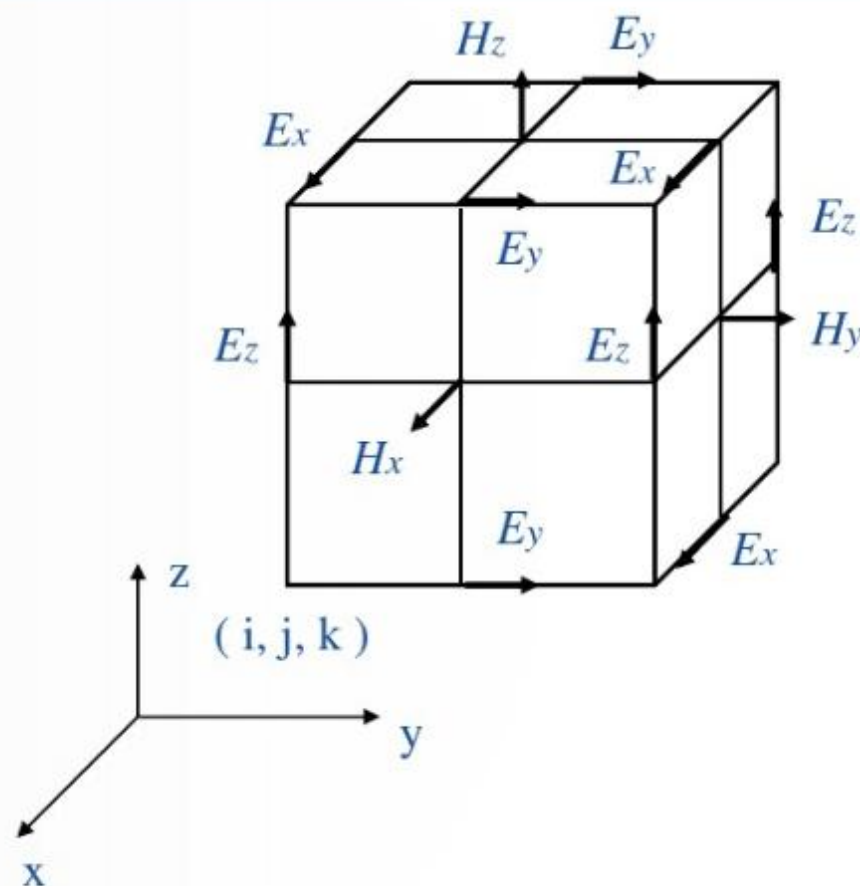


$$\tilde{H}_y^{n+1/2}(k + \frac{1}{2}) = \tilde{H}_y^{n-1/2}(k + \frac{1}{2}) - \frac{c\Delta t}{\Delta z \mu_r(k + \frac{1}{2})} [E_x^n(k+1) - E_x^n(k)]$$

$$E_x^{n+1}(k) = E_x^n(k) - \frac{c\Delta t}{\Delta z \epsilon_r(k)} \left[\tilde{H}_y^{n+1/2}(k + \frac{1}{2}) - \tilde{H}_y^{n+1/2}(k - \frac{1}{2}) \right]$$

三维Maxwell方程的Yee算法

1. 每一个磁场分量由四个电场分量环绕；
每一个电场分量由四个磁场分量环绕
2. 这种空间取样方式符合法拉第感应定律和安培环路定律
3. 电场和磁场在时间上交替抽样，抽样时间彼此相差半个时间步



FDTD离散中的Yee元胞

三维Maxwell方程的Yee算法

Maxwell旋度方程为

$$\nabla \times \vec{E} = -\rho \vec{H} - \mu \frac{\partial \vec{H}}{\partial t} \Rightarrow \begin{aligned} \frac{\partial H_x}{\partial t} &= \frac{1}{\mu} \left[\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} - \rho H_x \right] \\ \frac{\partial H_y}{\partial t} &= \frac{1}{\mu} \left[\frac{\partial E_z}{\partial x} - \frac{\partial E_x}{\partial z} - \rho H_y \right] \\ \frac{\partial H_z}{\partial t} &= \frac{1}{\mu} \left[\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} - \rho H_z \right] \end{aligned}$$

$$\nabla \times \vec{H} = \sigma \vec{E} + \varepsilon \frac{\partial \vec{E}}{\partial t} \Rightarrow \begin{aligned} \frac{\partial E_x}{\partial t} &= \frac{1}{\varepsilon} \left[\frac{\partial H_z}{\partial y} - \frac{\partial H_y}{\partial z} - \sigma E_x \right] \\ \frac{\partial E_y}{\partial t} &= \frac{1}{\varepsilon} \left[\frac{\partial H_x}{\partial z} - \frac{\partial H_z}{\partial x} - \sigma E_y \right] \\ \frac{\partial E_z}{\partial t} &= \frac{1}{\varepsilon} \left[\frac{\partial H_y}{\partial x} - \frac{\partial H_x}{\partial y} - \sigma E_z \right] \end{aligned}$$

三维Maxwell方程的Yee算法

采用时间平均近似

$$\frac{\partial H_x}{\partial t} = \frac{1}{\mu} \left[\frac{\partial E_y}{\partial z} - \frac{\partial E_z}{\partial y} - \rho H_x \right]$$

$$\left. \frac{\partial H_x}{\partial t} \right|_{i,j+\frac{1}{2},k+\frac{1}{2}}^n = \frac{H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} - H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n-\frac{1}{2}}}{\Delta t} + O[(\Delta t)^2]$$

$$H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^n = \frac{H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n+\frac{1}{2}} + H_x|_{i,j+\frac{1}{2},k+\frac{1}{2}}^{n-\frac{1}{2}}}{2} + O[(\Delta t)^2]$$

$$\left. \frac{\partial E_y}{\partial z} \right|_{i,j+\frac{1}{2},k+\frac{1}{2}}^n = \frac{E_y|_{i,j+\frac{1}{2},k+1}^n - E_y|_{i,j+\frac{1}{2},k}^n}{\Delta z} + O[(\Delta z)^2]$$

可以看到有个2阶小量，这个要忽略掉，因此是近似的

$$\left. \frac{\partial E_z}{\partial y} \right|_{i,j+\frac{1}{2},k+\frac{1}{2}}^n = \frac{E_z|_{i,j+1,k+\frac{1}{2}}^n - E_z|_{i,j,k+\frac{1}{2}}^n}{\Delta y} + O[(\Delta y)^2]$$

最后得：

$$H_x[i, j, k, n+1] = D_{aHx}[i, j, k] H_x[i, j, k, n] + D_{bHx}[i, j, k] \left\{ \frac{E_y[i, j, k+1, n] - E_y[i, j, k, n]}{\Delta z} - \frac{E_z[i, j+1, k, n] - E_z[i, j, k, n]}{\Delta y} \right\}$$

三维Maxwell方程的Yee算法

同理，可以得到其他2个磁场分量的FDTD方程

$$H_y[i, j, k, n+1] = D_{aHy}[i, j, k]H_y[i, j, k, n] + D_{bHy}[i, j, k] \left\{ \frac{E_z[i+1, j, k, n] - E_z[i, j, k, n]}{\Delta x} - \frac{E_x[i, j, k+1, n] - E_x[i, j, k, n]}{\Delta z} \right\}$$

$$H_z[i, j, k, n+1] = D_{aHz}[i, j, k]H_z[i, j, k, n] + D_{bHz}[i, j, k] \left\{ \frac{E_x[i, j+1, k, n] - E_x[i, j, k, n]}{\Delta y} - \frac{E_y[i+1, j, k, n] - E_y[i, j, k, n]}{\Delta x} \right\}$$

$$D_{aHw} = \frac{2\mu - \rho\Delta t}{2\mu + \rho\Delta t} \Big|_{H_w \text{ 所在空间位置}}$$

$$D_{bHw} = \frac{2\Delta t}{2\mu + \rho\Delta t} \Big|_{H_w \text{ 所在空间位置}}$$

三维Maxwell方程的Yee算法

利用对偶原理： $\vec{H} \rightarrow \vec{E}$, $E \rightarrow -\vec{H}$, $\mu \rightarrow \varepsilon$, $\rho \rightarrow \sigma$,

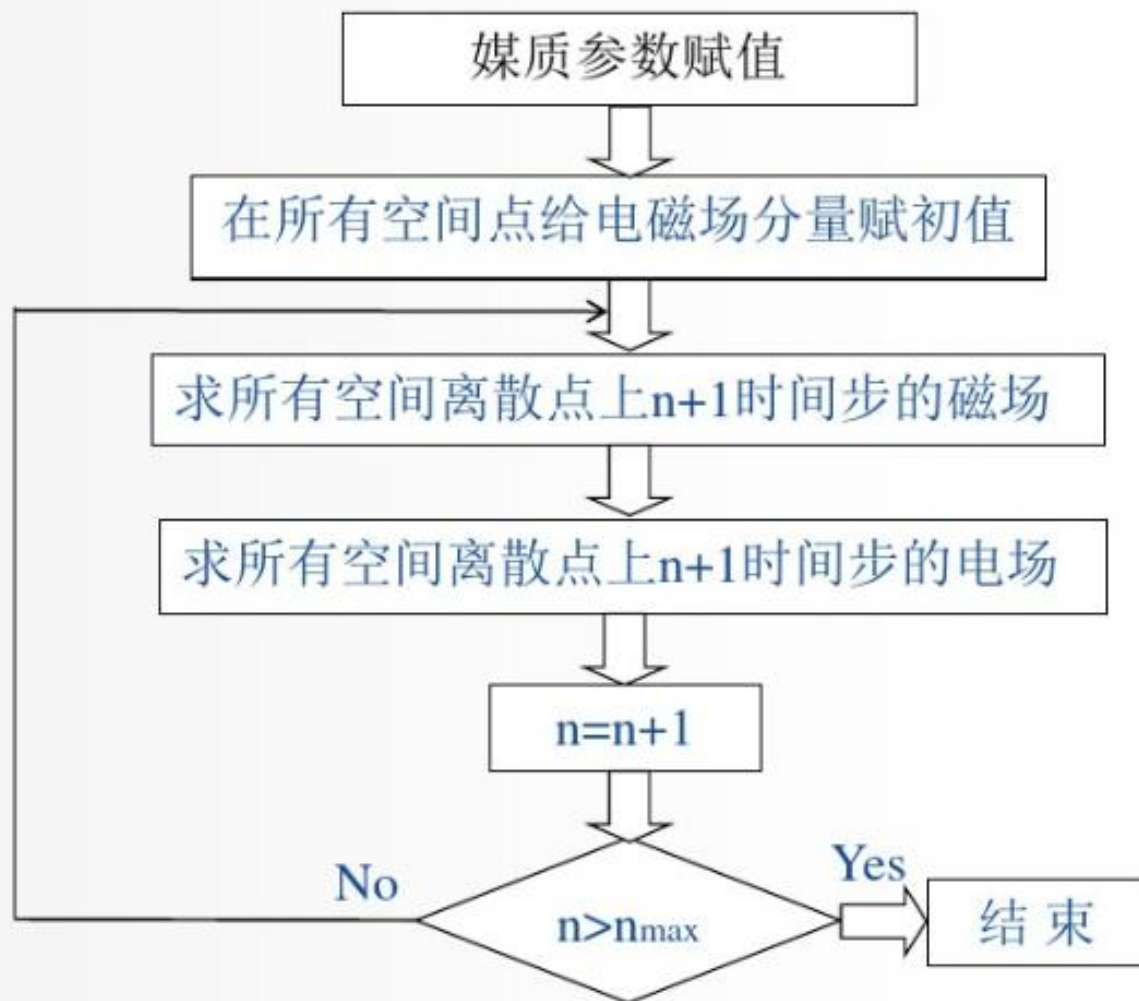
$D_a \rightarrow C_a, D_b \rightarrow C_b$ ，并注意到E与H在时间上差半个步长，
可以直接从磁场FDTD公式得到电场的FDTD公式。如：

$$E_x[i, j, k, n+1] = C_{aEx}[i, j, k]E_x[i, j, k, n] - C_{bEx}[i, j, k] \times \left\{ \frac{H_y[i, j, k+1, n+1] - H_y[i, j, k, n+1]}{\Delta z} - \frac{H_z[i, j+1, k, n+1] - H_z[i, j, k, n+1]}{\Delta y} \right\}$$

$$C_{aEw} = \frac{2\varepsilon - \sigma\Delta t}{2\varepsilon + \sigma\Delta t} \Big|_{E_w \text{ 所在空间位置}}$$

$$C_{bEw} = \frac{2\Delta t}{2\varepsilon + \sigma\Delta t} \Big|_{E_w \text{ 所在空间位置}}$$

三维Maxwell方程的Yee算法



三维Maxwell方程的Yee算法

介绍了求解矢量Maxwell方程的FDTD Yee算法，归纳起来，Yee算法的主要特点有：

- 1) Yee算法采用耦合的Maxwell旋度方程，同时在时间和空间求解电场和磁场，而不是采用波动方程只求解电场或磁场。
- 2) Yee网格在三维空间这样安排E和H分量，使得每一个E或H分量由四个H或E循环的分量所环绕。
- 3) Yee算法以蛙跳算法在时间上安排E和H分量。在某一时刻，使用前一时刻的E数据计算所有H分量。然后，再使用刚计算的H数据计算所有的E分量。如此循环，直至完成时间步进过程。

数值稳定性问题

(1) FDTD计算中每一步都是有误差的，随着时间步进，误差会不断积累。如果误差的积累不会造成总误差的增加，就成FDTD法是稳定的，否则成为不稳定的。数值不稳定性会造成计算结果随时间步进无限增加。

(2) FDTD法是有条件稳定的，即：时间步必须必须小于一定值以避免数值不稳定性。

(3) 数值稳定性分析方法是建立在Courant等人几十年前提出的经典方法基础上。这种方法首先把有限差分算法分解为相互分离的时间和空间本征值问题。

数值稳定性问题

二维的Yee算法数值Courant稳定性条件

$$\Delta t \leq \frac{1}{c \sqrt{\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2}}}$$

三维的Yee算法数值Courant稳定性条件

$$\Delta t \leq \frac{1}{c \sqrt{\frac{1}{(\Delta x)^2} + \frac{1}{(\Delta y)^2} + \frac{1}{(\Delta z)^2}}}$$

便于理解，当Yee元胞为立方体时，此两式表明时间间隔必须等于或小于波以光速通过yee元胞对角线长度1/3或1/2所需时间

差分近似所带来的数值色散

一维标量波动方程为例
$$\frac{\partial^2 u}{\partial t^2} = c^2 \frac{\partial^2 u}{\partial x^2}$$

设在离散空间点 (x_i, t_n) , 离散行波解为 $u_i^n = u(x_i, t_n) = e^{j(\omega n \Delta t - \tilde{k} i \Delta x)}$

将上式代入差分方程
$$u_i^{n+1} = \left(\frac{c\Delta t}{\Delta x}\right)^2 (u_{i+1}^n - 2u_i^n + u_{i-1}^n) + 2u_i^n - u_i^{n-1} + (\Delta t)^2 \{c^2 O[(\Delta x)^2] - O[(\Delta t)^2]\}$$

得：

$$e^{j\omega\Delta t} = \left(\frac{c\Delta t}{\Delta x}\right)^2 (e^{-j\tilde{k}\Delta x} - 2 + e^{j\tilde{k}\Delta x}) + (2 - e^{-j\omega\Delta t})$$

最后得色散关系

$$\tilde{k} = \frac{1}{\Delta x} \cos^{-1} \left\{ 1 + \left(\frac{\Delta x}{c\Delta t}\right)^2 [\cos(\omega\Delta t) - 1] \right\}$$

我们想要 k 与没加元胞之前的 k 一样，但是差分近似后 k 变的不一样

差分近似所带来的数值色散

- 情况1: 非常细网格 ($\Delta t \rightarrow 0, \Delta x \rightarrow 0$)

根据 $\cos x = 1 - x^2/2$, 当 $x \rightarrow 0$, 数值色散关系 (1-12) 变为

$$1 - \frac{(\omega \Delta t)^2}{2} \approx \left(\frac{c \Delta t}{\Delta x} \right)^2 \left(1 - \frac{(\tilde{k} \Delta x)^2}{2} - 1 \right) + 1$$

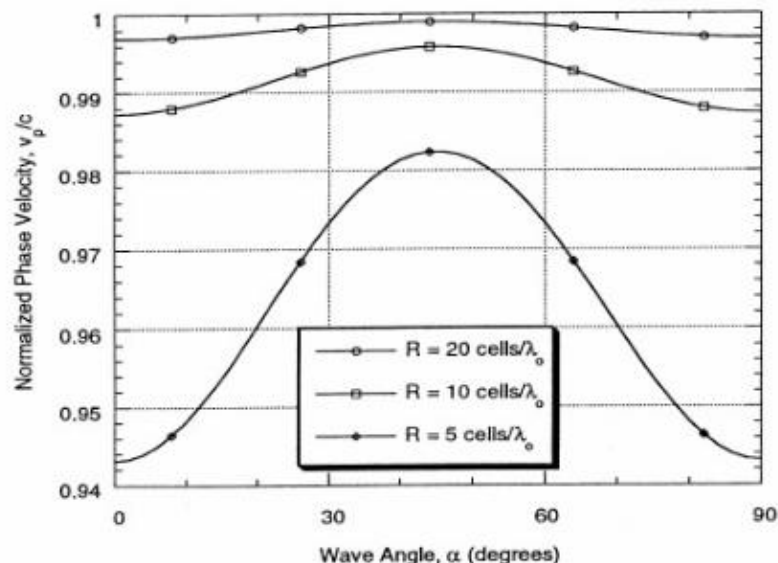
即, $\omega^2 \approx c^2 \tilde{k}^2$, 最后得 $\tilde{k} \approx \pm \frac{\omega}{c} = k$, 于是有 $\tilde{v}_p \approx v_p$ 。

所以, 在非常细的网格条件下, 差分解逼近精确解。

- 情况2: 魔时间步 ($c \Delta t = \Delta x$)

(1-12) 变为 $\cos(\omega \Delta t) = \cos(\tilde{k} \Delta x)$, 即 $\omega \Delta t = \pm \tilde{k} \Delta x$, $\tilde{k} = \pm \frac{\omega}{c} = k$ 。

所以, $\tilde{v}_p \approx v_p$ 。可见, 魔时间步下差分解与精确解相同。



将平面波带入差分方程所出现的稳定性、色散、以及各向异性, 这些特性并非由介质的物理特性所引起, 而是数值计算中的差分近似所致, 在FDTD数值计算中, 稳定性、色散、各向异性将影响计算精度。

吸收边界条件

理论上说，求解空间是无限大的，但是由于计算的数据容量问题，需要在有限空间的周围做特殊处理，使得向边界面行进的波在边界处保持“外向行进”，无明显的反射现象，并且不会使内部空间的场产生畸变。

- 1) Mur吸收边界条件
- 2) Berenger完美匹配层 (PML)
- 3) 各向异性匹配层吸收边界条件

Berenger完美匹配层 (PML)

只考虑二维TE情况，对于二维TM和三维情况可采用类似方法进行分析。分析范围在PML层内

$$\epsilon_0 \frac{\partial E_x}{\partial t} + \sigma E_x = \frac{\partial H_z}{\partial y}$$

$$\epsilon_0 \frac{\partial E_y}{\partial t} + \sigma E_y = -\frac{\partial H_z}{\partial x}$$

$$\mu_0 \frac{\partial H_z}{\partial t} + \sigma^* H_z = \frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x}$$

σ σ^*

分别表示自由空间中的电导率和磁阻率

Berenger为了引入规定损耗和阻抗匹配的新自由度，将 H_z 分裂为两个分量 H_{zx} 和 H_{zy} ，即 $H_z = H_{zx} + H_{zy}$ 同时引入了新的电导率 σ_x, σ_y 和磁损耗 σ_x^*, σ_y^* ，并规定TE情形的四个场分量（而不是通常的3个）由下列方程耦合在一起：

Berenger完美匹配层 (PML)

$$\varepsilon_0 \frac{\partial E_x}{\partial t} + \sigma_y E_x = \frac{\partial (H_{zx} + H_{zy})}{\partial y}$$

$$\varepsilon_0 \frac{\partial E_y}{\partial t} + \sigma_x E_y = -\frac{\partial (H_{zx} + H_{zy})}{\partial x}$$

$$\mu_0 \frac{\partial H_{zx}}{\partial t} + \sigma_x^* H_{zx} = -\frac{\partial E_y}{\partial x}$$

$$\mu_0 \frac{\partial H_{zy}}{\partial t} + \sigma_y^* H_{zy} = \frac{\partial E_x}{\partial y}$$

换句话说，Berenger构造了一般新的非物理媒质（称为PML媒质），在该媒质中场满足左端的方程（并不一定是Maxwell方程）

经过一系列公式推导：最后得出**无反射匹配条件**

$$\frac{\sigma_x}{\varepsilon_0} = \frac{\sigma_x^*}{\mu_0}$$

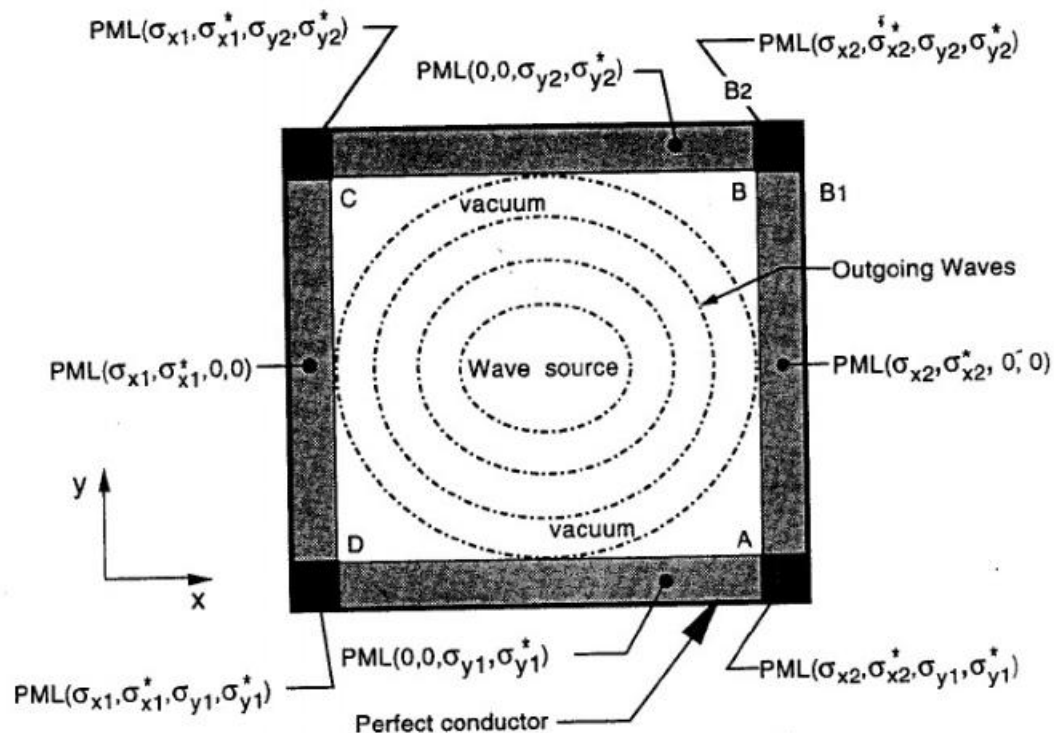
$$\frac{\sigma_y}{\varepsilon_0} = \frac{\sigma_y^*}{\mu_0}$$

表明PML媒质中的波阻抗与**入射角**无关，与真空中**波阻抗**相同，这意味着从真空中的**任意角度**入射到PML媒质交界面时将会**无反射**地进入PML媒质中，并在PML媒质中**衰减**地传播

$$Z = \frac{E_0}{H_0} = \sqrt{\frac{\mu_0}{\varepsilon_0}} \cdot G \quad G = \sqrt{W_x \cos^2 \phi + W_y \sin^2 \phi}$$

$$Z = \sqrt{\frac{\mu_0}{\varepsilon_0}}$$

Berenger完美匹配层 (PML)



1. 若分界面垂直于x轴（y轴），要求二者具有相同的横向电导率和磁导率，且横向和纵向电导率、磁导率均满足阻抗匹配条件。

Berenger完美匹配层 (PML)

如何计算：任何普通介质均可以视为特殊的PML介质，但麦克斯韦方程只需计算三个分量，而不是PML的四个分量。

另外由于在介质中电磁波衰减的很快，常规FDTD中Yee的差分格式已不再适用，需要指数差分

$$E_y \Big|_{i+1/2,j}^{n+1} = e^{-\sigma_x \Delta t / \varepsilon_0} E_y \Big|_{i+1/2,j}^n + \frac{1}{\sigma_x \Delta x} \left(e^{-\sigma_x \Delta t / \varepsilon_0} - 1 \right) \left(H_z \Big|_{i,j}^{n+\frac{1}{2}} - H_z \Big|_{i+1,j}^{n+\frac{1}{2}} \right)$$

$$E_x \Big|_{i,j+1/2}^{n+1} = e^{-\sigma_y \Delta t / \varepsilon_0} E_x \Big|_{i,j+1/2}^n + \frac{1}{\sigma_y \Delta y} \left(e^{-\sigma_y \Delta t / \varepsilon_0} - 1 \right) \left(H_z \Big|_{i,j+1}^{n+\frac{1}{2}} - H_z \Big|_{i,j}^{n+\frac{1}{2}} \right)$$

波源的设置

- ☞ 为了用FDTD法模拟电磁场工程问题，必须在FDTD网格中引入电磁波激励源。
- ☞ 常用的波源种类：
 - 平面波源—用于电磁散射问题；
 - 导波源—用于微波网络参数计算；
 - 电流源或电压源—用于微波电路或天线的激励。
- ☞ 常用的设置方式：
 - 初始条件；
 - 硬源；
 - 电流源；
 - 总场/散射场公式

硬波源

- ➡ 硬波源可以通过规定FDTD空间网格中电场或磁场分量满足所希望的时间函数进行简单地设置。如：在网格点 (i_s, j_s, k_s) 建立 E_z 的硬源为：

$$E_z \Big|_{i_s, j_s, k_s}^n = E_0 f(n\Delta t) \quad n \geq 0$$

- ➡ 硬波源相当于点源，模拟从源点向外辐射的具有与源函数性质相同的数值波。如果材料结构放在离波源一定距离的地方，辐射的数值波最终传播到这个结构，一部分传输过去，一部分反射。在原理上，时间步进可以连续进行，直到所有的瞬态波消失。

硬波源

常用源波形有以下三种

$$f(n\Delta t) = \begin{cases} \sin(2\pi f_0 n\Delta t) & \text{正弦波} \\ e^{-[(n-n_0)/n_{decay}]^2} & \text{Gaussian 脉冲} \\ e^{-[(n-n_0)/n_{decay}]^2} \sin(2\pi f_0 (n-n_0)\Delta t) & \text{带通Gaussian 脉冲} \end{cases}$$

正弦波源为时谐源，提供了频率为 f_0 的正弦波。

FDTD模拟时，一般采用Gaussian脉冲信号激励，因为它可以提供宽频带特性。Gaussian脉冲信号的频谱也是Gaussian函数。

$$G(\omega) = \sqrt{\pi} n_{decay} \Delta t \exp\left[-(n_{decay}^2 \Delta t^2 \omega^2 / 4)\right]$$

硬波源

但是，在连续时间步进时，当反射波到达波源位置，由于**波源处总场**已被规定，并没有考虑网格中可能的反射波（所以称为硬源），硬源对这些反射波就会造成寄生的非物理的**再次反射**。当规定了一个表面上的总场而没有考虑反射场值时，寄生反射就一定会发生。

解决这一问题的一个简单方法是当反射波到达波源位置时**去掉波源**。这就要求源离散射体**足够远**，使散射波到达源位置时源入射波**已基本为零**，所以这种方法不适合正弦入射波源。

FDTD Lumerical 教程

<https://www.eeo.cn/live.php?lessonKey=67edfcce894ee630>

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